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# An Adjoint Sensitivity Model for Radionuclide Transport in Heterogeneous Discrete Fractured Porous Formation in Steady-State Regime

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**Key Points:**

- A full mathematical/numerical adjoint sensitivity model for steady-state RN transport in heterogeneous discrete fractured media is presented
- Analytical solutions for concentrations and adjoint states have been derived for the case of layered formations including a fracture
- The developed adjoint approach has been coupled with the differential sensitivity analysis method and applied to a real field case

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**Abstract** The development of the adjoint state method for the system of partial differential equations (PDEs) describing radionuclide transport in heterogeneous discrete fractured porous media in steady-state regime is presented in this study for both continuous and discrete adjoint approaches. The continuous adjoint approach has its own advantage in the sense that the adjoint system can be derived in terms of PDEs which can be solved analytically for specific performance measures. These PDEs can then be used to obtain a physical interpretation of the adjoint state functions. The discrete adjoint approach permits a simpler mathematical treatment based on the use of the same matrix (or its transpose) of the linear system of the discretized transport equations referring to from hereon as the forward problem. Analytical solutions for both the forward problem and its associated adjoint problem have been derived for radionuclide transport in layered formations including a fracture using an analytical inversion of tridiagonal matrices. A set of 14 formulas of sensitivity coefficients with respect to pertinent system parameters has been derived and presented in integral forms. Analytical results have been compared with numerical results and have shown excellent agreements. The discrete adjoint approach has been coupled with the differential sensitivity analysis method and applied to a real field case involving the performance assessment of a site currently being considered for underground nuclear storage in northern Switzerland, Zürich Nordost. The results show that when a fracture is present in the vicinity of waste repository, the state parameter controlling flow through the fracture, referred to herein as fracture transmissivity, is among the important parameters which should be considered in sensitivity analysis studies.

## 1. Introduction

Nuclear waste programs have investigated the feasibility of siting repositories in deep formations to ensure the safe long-term storage of radioactive waste. The ultimate decision to construct and operate a repository is made after many years of site characterization and performance assessment activities. Decision makers rely on models to ascertain the expected performance of the geosystem as a barrier to offsite transport of radionuclides (RNs). Given the reality that the near-field and far-field hydraulic properties which describe the geosystem can never be known with certainty, the propagation of parameter uncertainty to the safety calculations is imperative in order to inform decision makers of the risk of a site under consideration. Nuclear waste isolation programs in the United States (Waste Isolation Pilot Plant (WIPP), https://www.wipp.energy.gov), Canada (Canadian Nuclear Safety Commission, https://nuclearsafety.gc.ca), Europe (France (ANDRA: https://www.andra.fr), Belgium (ONDRAF: https://www.ondraf.be), Switzerland (NAGRA: https://www.nagra.ch), United Kingdom (Radioactive Waste Management: https://rwm.nda.gov.uk)) and the Nordic Countries (Sweden (SKB: https://www.skb.com), Finland (Posiva Oy: https://www.posiva.fi)) have characterized sites in a variety of geologic formations with low permeability host rock, for example, clay, salt, granite, and volcanic tuff. The presence of fractures in a host rock is an important factor for the movement of RNs by diffusion and advection and therefore to the performance of the host rock as a geologic barrier to offsite transport. Moreover, fractures and/or micro and macro cracks can also develop in concrete structures which line waste repositories as a result of interactive physical, chemical, and mechanical processes (Neville, 1995; Page & Page, 2007; Seetharam et al., 2014). The effect of fractures on the release of RNs depends on interactions between fracture characteristics, porous matrix properties, and location of radioactive wastes. Fractures may accelerate the release of RNs from the subsurface waste repositories to the overburden due to their high permeability and porosity compared to those of the surrounding rock formations. On the other hand, exchange between fracture and matrix allows the diffusion of RNs from

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the fracture to the porous matrix which generally has high sorption capacity. Therefore, sensitivities of user-defined measures of system behavior (called also performance measures) with respect to fracture and matrix parameters (mainly fracture transmissivity, effective diffusion coefficients and sorption coefficients of matrix) are important for obtaining an enhanced understanding of the fractured system performance.

The change of a model results (i.e., performance measures) due to the variation of an input parameter can be measured using the sensitivity coefficients. Sensitivities can be expressed mathematically as the partial derivatives of the performance measure with respect to the parameter. They are considered the main tool of any sensitivity analysis method. In the context of modeling RN release from waste repositories, sensitivity coefficients allow the identification of the relevant flow and transport processes and thus lead to a better understanding of the geosystem's performance and ultimately the safety of a potential repository.

Sensitivity coefficients can be obtained by various methods. The influence coefficient method (known also as the perturbation method or the finite difference approximation (FDA) method) and the sensitivity equation method are the most used methods in subsurface hydrology (Kool & Parker, 1988; Kool et al., 1987; Yeh, 1986). The FDA approach is frequently used in numerical simulators for calculating the Jacobian matrix required in inverse modeling and optimization (e.g., the iTOUGH2 simulator (Finsterle, 2007) and the PEST++ software (White et al., 2020)). The method is easy to use as it does not require any special implementation. It uses the concept of parameter perturbation. Each parameter is perturbed by a small amount and one forward model run is needed to solve using the perturbed parameter value while the other parameters are kept to their original values. The sensitivity coefficient is then approximated using finite differences. This procedure is repeated for all parameters, thus if there are $N_p$ parameters to be identified, the governing equation has to be solved $(N_p + 1)$ times. In the sensitivity equation method, a set of sensitivity equations are obtained by taking the partial derivatives of the governing equations and their initial and boundary conditions with respect to each parameter. The set of sensitivity equations can be solved by the same method used for solving the forward problem. The solutions of the sensitivity equations (known as the state sensitivities) are then used to compute the sensitivity coefficients. The number of simulation runs required to generate the sensitivity coefficients per iteration is $(N_p + 1)$, which is the same as that of the influence coefficient method. Therefore, these methods are not suitable for problems with large number of parameters due to their computational cost.

In the early seventies, Chavent (1974) introduced the adjoint state method to the theory of inverse problems to efficiently compute the gradients of a functional with respect to the parameters of the system (see also Sun, 1994). This adjoint approach which originates from control theory (Lions & Magenes, 1972) aims to overcome the problem of the computational cost associated with the use of the influence coefficient method and the sensitivity equation method. The adjoint approach is computationally attractive because only one extra system (called the adjoint state system) needs to be solved. This cost is independent of the number of model parameters and depends only on the performance measure. Therefore, the adjoint state approach is attractive when the number of parameters is much larger than the number of performance measures. The main difficulty of the adjoint methodology is that it can be very tedious and time-consuming to implement because it requires detailed manipulations involving the forward model equations. However, the method provides exact sensitivities in contrast to other numerical approaches (i.e., the influence coefficient method). Recently, the adjoint state method has been applied and used successfully in problems related to flow and transport processes in porous media (see for instance Delay et al., 2017, 2019; Hayek et al., 2019, 2020; Lu & Vesselinov, 2015).

In the framework of radioactive waste disposal programs, numerical models describing RN release from waste repositories usually involve a large number of physical parameters which includes diffusion coefficients, sorption coefficients, porosities, and hydraulic conductivities of various geological formations. The computation of the sensitivities of the output of the model (e.g., the RN flux at a given interface or the mass of RNs in a given region) with respect to the set of model parameters needs to be evaluated by applying efficient techniques such as the adjoint state method.

The purpose of the present work is to develop the adjoint state methodology to the problem of RN transport in heterogeneous discrete fractured porous media in steady-state regime. Both continuous and discrete adjoint approaches have been employed. For the continuous approach, analytical solutions of RN transport in layered formations have been developed for both forward and adjoint systems. The adjoint approach has

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been coupled with the differential sensitivity analysis method and applied to a real field case represented by the siting region Zürich Nordost in northern Switzerland.

## 2. The Mathematical Model

### 2.1. The 2D Steady-State Transport Equation

The partial differential equation (PDE) governing the steady-state transport of a decaying RN in a single heterogenous porous formation by two-dimensional dispersion and one-dimensional advection is

$$\frac{\partial}{\partial x}\left(-D_{e,x}\frac{\partial C}{\partial x}\right)+\frac{\partial}{\partial z}\left(-D_{e,z}\frac{\partial C}{\partial z}\right)+\frac{\partial}{\partial x}\left(VC\right)+\lambda\phi RC+Q=0 \tag{1}$$

where $x$ and $z$ are the space coordinates ($L$) with $x$ represents the direction of groundwater flow, $\phi$ is the porosity of the porous formation (–), $R$ is the RN retardation factor due to sorption (–), $V$ is the Darcy velocity ($LT^{-1}$), $D_{e,x}$ and $D_{e,z}$ are the "effective" hydrodynamic dispersion coefficients in the $x$ and $z$ directions ($L^2T^{-1}$), respectively, $C$ is the RN concentration ($ML^{-3}$), and $Q$ is the volumetric sink/source term ($ML^{-3}T^{-1}$). For the general heterogeneous case, the parameters of the system (i.e., porosity, retardation factor, fluid flow velocity, and dispersion coefficients) may be dependent of the space coordinates.

The Darcy velocity is related to the groundwater fluid flow velocity $u$ through the porosity

$$V=\phi u \tag{2}$$

The effective hydrodynamic dispersion coefficients $D_{e,x}$ and $D_{e,z}$ represent the process of dispersion due to both fluid flow velocity and molecular diffusion. With the assumption of flow in the direction of the $x$ axis, the effective hydrodynamic dispersion coefficients are defined by

$$D_{e,x}=\alpha_L|V|+d_{e,x} \tag{3}$$

$$D_{e,z}=\alpha_T|V|+d_{e,z} \tag{4}$$

where $\alpha_L$ and $\alpha_T$ are the longitudinal and transverse dispersivity coefficients ($L$), respectively, and $d_{e,x}$ and $d_{e,z}$ are the effective molecular diffusion coefficients ($L^2T^{-1}$) of the RN in both $x$ and $z$ directions, respectively. The effective molecular diffusion coefficients are defined by (Boving & Grathwohl, 2001)

$$d_{e,x}=\frac{d_{\text{aq}}\phi\delta}{\tau_x} \tag{5}$$

$$d_{e,z}=\frac{d_{\text{aq}}\phi\delta}{\tau_z} \tag{6}$$

where $d_{\text{aq}}$ is the aqueous diffusion coefficient in pure water ($L^2/T$), $\tau_x$ and $\tau_z$ are the tortuosity factors ($\geq 1$) in the $x$ and $z$ directions, respectively, which are geometric parameters and account for the pore space geometry ($\tau_x=\tau_z$ if the total pore space is isotropic), and $\delta$ is dimensionless constrictivity ($\leq 1$) which is only important if the size of the solute becomes comparable to the size of the pore (see Boving & Grathwohl, 2001 and Shackelford, 1991 for details). The substitution of Equations 2, 5 and 6 into Equations 3 and 4 gives

$$D_{e,x}=\phi D_x \tag{7}$$

$$D_{e,z}=\phi D_z \tag{8}$$

where $D_x=\alpha_L|u|+d_x$ and $D_z=\alpha_T|u|+d_z$ are dispersion coefficients written in terms of the fluid flow velocity $u$ and the molecular diffusion coefficients $d_x$ and $d_z$ (scaled by the tortuosity factors).

### 2.2. Steady-State Transport Equations in Fractured Medium

In fractured porous media, Equation 1 can be used to describe transport of RN in both fracture and porous matrix, with values of dispersion coefficients, groundwater flow velocities, porosities, and retardation factors dependent on each region. We shall use the superscripts $f$ and $m$ on these parameters to signify their values in the fracture and matrix, respectively.

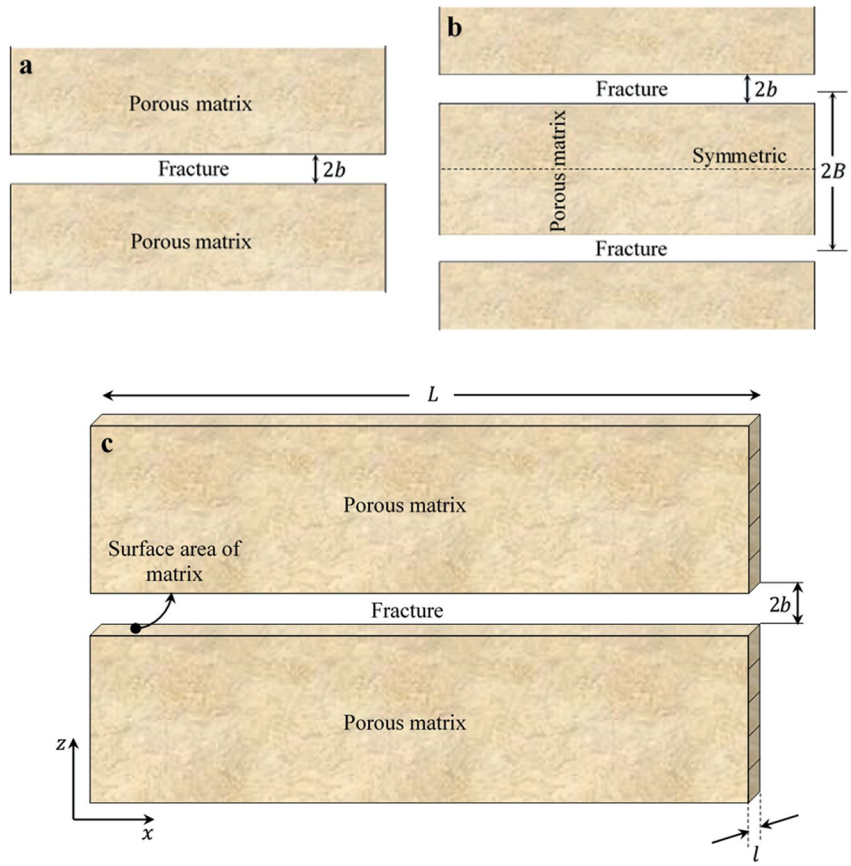



**Figure 1.** Configuration of fracture-matrix systems: (a) single fracture system, (b) system of parallel fractures, (c) 3D representation showing the matrix surface area.

The axis passing through the middle of the fracture is a symmetrical axis of the fracture-matrix geometry (see Figures 1 and 2). Therefore, we need to consider only one-half of a fracture (i.e., $0 \le x \le L$ and $0 \le z \le b$) and one half of the intervening porous matrix (i.e., $0 \le x \le L$ and $b \le z \le B$), where $b$ is the fracture aperture $(L)$, and $2B$ is fracture spacing $(L)$. We shall denote by $\Omega^f = [0, L] \times [0, b]$ and $\Omega^m = [0, L] \times [b, B]$ the domains of definition of the fracture and the matrix, respectively.

We consider the case of thin rigid fractures extending across a saturated porous rock as shown on Figure 1. Figure 1a displays the case of a porous medium containing a single fracture of length $L$ while Figure 1b shows the case of a porous medium containing an infinite number of equally spaced identical fractures. It is assumed that the width of the fracture $2b$ $(L)$ is much smaller than its length $L$ so that the transport of RN through the fractures is mainly 1-D in the direction of the groundwater flow. Note that we need to consider only one half of a fracture and one half of the intervening porous matrix because the fracture-matrix geometry repeats itself (see Figure 1).

Because of the flow along the thin fractures is 1-D, the differential equation describing the RN transport along a fracture can be written in a similar form as (Equation 1) but without the second term accounting for diffusion along the axis which is perpendicular to the groundwater flow direction. This equation is connected to the transport equation of the matrix by a diffusive loss term $\phi^f \delta^f F$ $(ML^{-2}T^{-1})$ which represents the diffusive loss from the fracture to the porous matrix:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(-D_e^f \frac{\mathrm{d}C^f}{\mathrm{d}x}\right) + \frac{\mathrm{d}}{\mathrm{d}x}\left(V^f C^f\right) + \lambda \phi^f R^f C^f + \phi^f \delta^f F + Q_f = 0 \tag{9}$$

where $\delta^f$ is the surface area of matrix per unit volume of flowing water $(L^{-1})$ which can be written as follows (see Figure 1c)



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$$\delta^f = \frac{L \times l}{\phi^f \left(L \times l \times b\right)} = \frac{1}{\phi^f b} \tag{10}$$

with $l$ represents the width of the fractured medium ($L$) which is assumed to be equal to 1 in practical applications, and $F$ is the loss rate from the fracture to the matrix ($ML^{-2}T^{-1}$) defined by

$$F = -D^m_{e,z} \frac{\partial C^m}{\partial z}\Big|_{z=b} \tag{11}$$

The loss rate from the fracture to the matrix as defined by (Equation 11) is valid for fractures which are parallel to the groundwater flow direction. These fractures are then represented as thin layers crossing the porous matrix as shown in Figure 1. In a system of fractured layered geological formations, the fracture intersects normally to the rock layers and is of varying aperture across its length (Gureghian et al., 1993; Wu et al., 1993). Therefore, the fracture network is serial in nature and of uniform thickness with each layer as shown in Figure 2.

The substitution of Equations 10 and 11 into Equation 9 gives the transport equation in the fracture in the following form (for $(x, z) \in \Omega^f$):

$$L^f \equiv \frac{\mathrm{d}}{\mathrm{d}x}\left(-D^f_e \frac{\mathrm{d}C^f}{\mathrm{d}x}\right) + \frac{\mathrm{d}}{\mathrm{d}x}\left(V^f C^f\right) + \lambda \phi^f R^f C^f - \frac{D^m_{e,z}}{b}\frac{\partial C^m}{\partial z}\Big|_{z=b} + Q^f = 0 \tag{12}$$

where $L^f$ denotes the linear transport operator associated with the fracture.

The differential equation describing the transport of RN through the porous matrix is similar to Equation 1 which can be written (using the superscript $m$) for $(x, z) \in \Omega^m$ as follows

$$L^m \equiv \frac{\partial}{\partial x}\left(-D^m_{e,x}\frac{\partial C^m}{\partial x}\right) + \frac{\partial}{\partial z}\left(-D^m_{e,z}\frac{\partial C^m}{\partial z}\right) + \frac{\partial}{\partial x}\left(V^m C^m\right) + \lambda \phi^m R^m C^m + Q^m = 0 \tag{13}$$

In Equation 13, $L^m$ is the linear transport operator associated with the matrix.

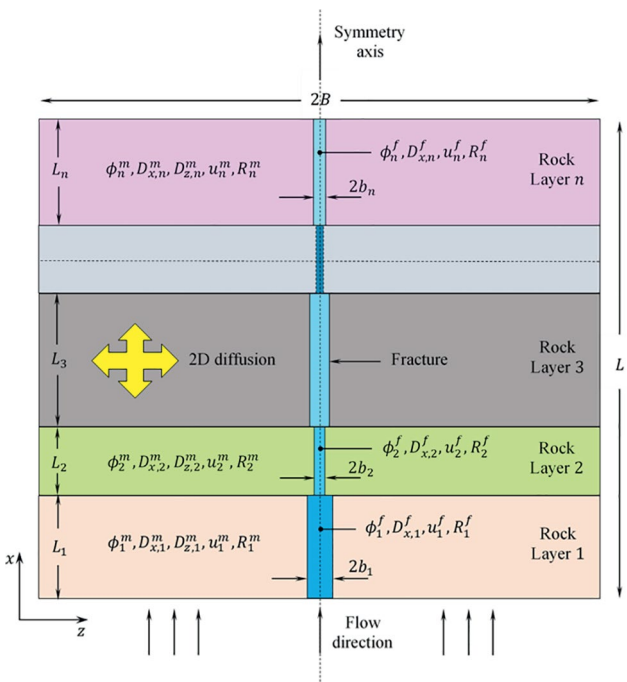



**Figure 2.** Vertical cross section of geological formation showing the migration pathways in a system of horizontal parallel homogeneous layers of rock intersected by a vertical fracture of varying aperture ($x$ denotes the vertical axis parallel to the flow direction and $z$ denotes the horizontal axis perpendicular to the flow direction).

### 2.3. Boundary Conditions

The set of differential Equations 12 and 13 is subject to the following boundary conditions:

For the fracture:

$$C^f\left(0\right) = C_0^f, \quad 0 \le z \le b \tag{14}$$

$$C^f\left(L\right) = C_L^f, \quad 0 \le z \le b \tag{15}$$

For the matrix:

$$C^m\left(0, z\right) = C_0^m, \quad b \le z \le B \tag{16}$$

$$C^m\left(L, z\right) = C_L^m, \quad b \le z \le B \tag{17}$$

$$C^m\left(x, b\right) = C^f\left(x\right), \quad 0 \le x \le L \tag{18}$$

$$\frac{\partial C^m}{\partial z}\Big|_{z=B} = 0, \quad 0 \le x \le L \tag{19}$$

where $C_0^f$ and $C_L^f$ (resp. $C_0^m$ and $C_L^m$) are given prescribed concentrations ($ML^{-3}$) associated with the fracture (resp. the matrix), respectively. The condition (Equation 18) insures the continuity of the RN concentration at the interface between the fracture and the matrix.



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### 2.4. Performance Measure and Sensitivity Coefficient

In sensitivity analysis, a response function or performance measure is a scalar function of the outputs (i.e., RN concentrations in the fracture and the matrix). For discrete fractured media, the outputs $C^f$ and $C^m$ are defined over two adjacent domains: fracture and matrix. For the steady-state case studied herein, the general form of any performance measure can be expressed as follows:

$$J\left(p, C^f, C^m\right) = \iint_{\Omega^f} G^f\left(p, C^f\right) d\Omega + \iint_{\Omega^m} G^m\left(p, C^m\right) d\Omega \tag{20}$$

where $p$ is a system parameter, $G^f$ and $G^m$ are unspecified functions of the system states ($C^f$ and $C^m$) associated with the fracture and the matrix, respectively, and $d\Omega = \mathrm{d}x\mathrm{d}z$.

Note that since the fluid flow in the fracture is modeled as 1-D linear flow, the first integral of the right-hand side of (Equation 20) can be simplified to $\int_0^L bG^f(p, C^f)\mathrm{d}x$. However, we shall use the more general expression (Equation 20) through this paper.

The marginal sensitivity coefficients of the performance measure $J$ to any parameter $p$ can be obtained by taking the derivative of (Equation 20) with respect to $p$. Bearing in mind that $b$ might be one of the system parameters and that, $\iint_{\Omega^f} d\Omega = \int_0^L \int_0^b \cdot d\Omega$ and $\iint_{\Omega^m} d\Omega = \int_0^L \int_b^B \cdot d\Omega$, the derivative of (Equation 20) with respect to $p$ can be obtained as follows after using the Leibniz integral rule:

$$\begin{aligned}\frac{\partial J}{\partial p} &= \iint_{\Omega^f}\left[\frac{\partial G^f\left(p, C^f\right)}{\partial p} + \frac{\partial G^f\left(p, C^f\right)}{\partial C^f}\psi^f\right] d\Omega \\ &+ \iint_{\Omega^f}\frac{\partial b}{\partial p}\frac{1}{b}\left[G^f\left(b, C^f\right) - G^m\left(b, C^f\right)\right] d\Omega \\ &+ \iint_{\Omega^m}\left[\frac{\partial G^m\left(p, C^m\right)}{\partial p} + \frac{\partial G^m\left(p, C^m\right)}{\partial C^m}\psi^m\right] d\Omega\end{aligned} \tag{21}$$

where $\psi^f = \dfrac{\partial C^f}{\partial p}$ and $\psi^m = \dfrac{\partial C^m}{\partial p}$ are the state sensitivities associated with the fracture and the matrix, respectively. The functions $\psi^f$ and $\psi^m$ are (as $C^f$ and $C^m$) defined over the sub-domains $\Omega^f$ and $\Omega^m$, respectively. In the second integral of the right-hand side of (Equation 21) we have used condition (Equation 18) so that $G^m(b, C^m) = G^m(b, C^f)$.

The terms $\partial G^f(p, C^f)/\partial p$ and $\partial G^m(p, C^m)/\partial p$ in (Equation 21) represent the explicit dependence of $J$ on the parameter $p$. These terms can be calculated analytically and are called the direct effects. The terms containing $\psi^f$ and $\psi^m$ represent the indirect effects due to the implicit dependence of $J$ on $p$ through the RN concentrations $C^f$ and $C^m$, respectively. These terms are more complicated to calculate because they require the determination of the state sensitivities $\psi^f$ and $\psi^m$, respectively. The second integral of the right-hand side of (21) is zero except when $p = b$ (i.e., $\partial b/\partial p = 1$ if $p = b$ and $\partial b/\partial p = 0$ if $p \neq b$).

The state sensitivities can be computed by taking the derivatives $\partial L^f/\partial p$ and $\partial L^m/\partial p$ of the transport operators which give Equations A1 and A2 of Appendix A after using Equations 12 and 13, respectively. The associated boundary conditions can be easily obtained by taking the derivatives with respect to $p$ of Equations 14–19 (see Equations A3–A8 of Appendix A). The system of Equations A1 and A2 with boundary conditions Equations A3–A8 can be solved numerically for each parameter $p$ to obtain $\psi^f$ and $\psi^m$, respectively. However, the resolution requires one run per parameter number so that the number of simulations is proportional to the number of parameters. This becomes cumbersome and computationally expensive when the number of parameters is large. An alternative solution is to use the adjoint state method which is computationally more efficient.

## 3. The Adjoint State Method

The adjoint state method requires only one additional system of equations to be solved for each performance measure. This system is called the adjoint state system and its resolution does not depend on the number of parameters. The adjoint system together with the sensitivity coefficients can be obtained at once

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as shall be seen below. There exist two main approaches for the derivation of the adjoint state equations and sensitivity coefficients for a given numerical model: the continuous approach and the discrete approach. With the continuous approach, the adjoint state equations are formulated as PDEs starting from the equations representing the primary problem Equations 12 and 13. Therefore, the sensitivity coefficients are presented in integral form. The discrete approach consists in using the discretized matrix equations form of the primary problem (which is obtained after using a numerical scheme) to derive an associated discrete matrix equation of adjoint states. Continuous and discrete approaches are both used in the present work and they are presented below.

### 3.1. Continuous Approach

The starting point of this approach is the primary problem defined by the set of Equations 12 and 13. For a given parameter $p$, the approach consists in differentiating Equations 12 and 13 with respect to $p$ as shown in Equations A1 and A2 in Appendix A. Then, by multiplying $\partial L^f/\partial p$ and $\partial L^m/\partial p$ by arbitrary functions $\psi_*^f$ and $\psi_*^m$ defined over $\Omega^f$ and $\Omega^m$, respectively, and integrating over the corresponding domains, we get

$$\iint_{\Omega^f} \frac{\partial L^f}{\partial p}\psi_*^f d\Omega = 0 \tag{22}$$

and

$$\iint_{\Omega^m} \frac{\partial L^m}{\partial p}\psi_*^m d\Omega = 0 \tag{23}$$

Adding the terms on the left side of (22) and (23) to the right side of (21), the marginal sensitivity becomes

$$\begin{aligned}\frac{\partial J}{\partial p} = & \iint_{\Omega^f}\left\{\frac{\partial G^f\left(p,C^f\right)}{\partial p}+\frac{\partial G^f\left(p,C^f\right)}{\partial C^f}\psi^f+\frac{\partial L^f}{\partial p}\psi_*^f\right.\\ & \left.+\frac{\partial b}{\partial p}\frac{1}{b}\left[G^f\left(b,C^f\right)-G^m\left(b,C^f\right)\right]\right\}d\Omega\\ & +\iint_{\Omega^m}\left[\frac{\partial G^m\left(p,C^m\right)}{\partial p}+\frac{\partial G^m\left(p,C^m\right)}{\partial C^m}\psi^m+\frac{\partial L^m}{\partial p}\psi_*^m\right]d\Omega\end{aligned} \tag{24}$$

In Equation 24, $\partial L^f / \partial p$ and $\partial L^m / \partial p$ contain the state sensitivities and their derivatives (see Equations A1 and A2 of Appendix A). These terms are unknowns and we don't want to calculate them. The idea is to use the partial integration rule to obtain only terms multiplied with $\psi^f$ and $\psi^m$ (and not their derivatives). Doing so, we get after some mathematical manipulations (see Appendix A for the detailed mathematical derivation)

$$\begin{aligned}\frac{\partial J}{\partial p} = & \iint_{\Omega^f} A^f\psi^f d\Omega+\iint_{\Omega^m} A^m\psi^m d\Omega\\ & +\iint_{\Omega^f}\left\{+\frac{\partial G^f}{\partial p}+\frac{\partial \phi^f}{\partial p}\lambda R^f C^f\psi_*^f+\frac{\partial R^f}{\partial p}\lambda\phi^f C^f\psi_*^f+\frac{\partial D_e^f}{\partial p}\frac{\mathrm{d}C^f}{\mathrm{d}x}\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\right.\\ & -\frac{\partial V^f}{\partial p}C^f\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}+\frac{\partial C_L^f}{\partial p}\frac{D_e^f}{L}\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\Big|_{x=L}-\frac{\partial C_0^f}{\partial p}\frac{D_e^f}{L}\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\Big|_{x=0}\\ & +\frac{\partial b}{\partial p}\frac{1}{b}\left[D_{e,z}\left(\frac{1}{b}\frac{\partial C^m}{\partial z}\Big|_{z=b}-\frac{\partial^2 C^m}{\partial z^2}\Big|_{z=b}\right)\psi_*^f+G^f\left(b,C^f\right)\right.\\ & \left.\left.-G^m\left(b,C^f\right)\right]+\frac{\partial Q^f}{\partial p}\psi_*^f\right\}d\Omega\\ & +\iint_{\Omega^m}\left\{\frac{\partial G^m}{\partial p}+\frac{\partial \phi^m}{\partial p}\lambda R^m C^m\psi_*^m+\frac{\partial R^m}{\partial p}\lambda\phi^m C^m\psi_*^m\right.\\ & +\frac{\partial D_{e,x}}{\partial p}\frac{\partial C^m}{\partial x}\frac{\partial \psi_*^m}{\partial x}+\frac{\partial D_{e,z}}{\partial p}\frac{\partial C^m}{\partial z}\frac{\partial \psi_*^m}{\partial z}-\frac{\partial V^m}{\partial p}C^m\frac{\partial \psi_*^m}{\partial x}\\ & \left.+\frac{\partial C_L^m}{\partial p}\frac{D_{e,x}}{L}\frac{\partial \psi_*^m}{\partial x}\Big|_{x=L}-\frac{\partial C_0^m}{\partial p}\frac{D_{e,x}}{L}\frac{\partial \psi_*^m}{\partial x}\Big|_{x=0}+\frac{\partial Q^m}{\partial p}\psi_*^m\right\}d\Omega\end{aligned} \tag{25}$$

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**Table 1**
*General Formulas of the Marginal Sensitivity Coefficients Associated With Each Parameter*

| Parameter $p_j$ | Sensitivity coefficient $\partial J/\partial p_j$ over $\Omega_j^f \subseteq \Omega^f$ or $\Omega_j^m \subseteq \Omega^m$ | N° |
|---|---|---|
| $\phi_j^f$ | $\iint_{\Omega_j^f}\left(\frac{\partial G^f}{\partial \phi_j^f}+\lambda R^f C^f \psi_*^f+\frac{\partial Q^f}{\partial \phi_j^f}\psi_*^f\right)d\Omega$ | (I) |
| $R_j^f$ | $\iint_{\Omega_j^f}\left(\frac{\partial G^f}{\partial R_j^f}+\lambda \phi^f C^f \psi_*^f+\frac{\partial Q^f}{\partial R_j^f}\psi_*^f\right)d\Omega$ | (II) |
| $V_j^f$ | $\iint_{\Omega_j^f}\left(\frac{\partial G^f}{\partial V_j^f}-C^f\frac{d\psi_*^f}{dx}+\frac{\partial Q^f}{\partial V_j^f}\psi_*^f\right)d\Omega$ | (III) |
| $D_{e,j}^f$ | $\iint_{\Omega_j^f}\left(\frac{\partial G^f}{\partial D_{e,j}^f}+\frac{dC^f}{dx}\frac{d\psi_*^f}{dx}+\frac{\partial Q^f}{\partial D_{e,j}^f}\psi_*^f\right)d\Omega$ | (IV) |
| $C_{0,j}^f$ | $\iint_{\Omega_j^f}\left(\frac{\partial G^f}{\partial C_{0,j}^f}-\frac{D_e^f}{L}\frac{d\psi_*^f}{dx}\Big\vert_{x=0}+\frac{\partial Q^f}{\partial C_{0,j}^f}\psi_*^f\right)d\Omega$ | (V) |
| $C_{L,j}^f$ | $\iint_{\Omega_j^f}\left(\frac{\partial G^f}{\partial C_{L,j}^f}-\frac{D_e^f}{L}\frac{d\psi_*^f}{dx}\Big\vert_{x=L}+\frac{\partial Q^f}{\partial C_{L,j}^f}\psi_*^f\right)d\Omega$ | (VI) |
| $\phi_j^m$ | $\iint_{\Omega_j^m}\left(\frac{\partial G^m}{\partial \phi_j^m}+\lambda R^m C^m \psi_*^m+\frac{\partial Q^m}{\partial \phi_j^m}\psi_*^m\right)d\Omega$ | (VII) |
| $R_j^m$ | $\iint_{\Omega_j^m}\left(\frac{\partial G^m}{\partial R_j^m}+\lambda \phi^m C^m \psi_*^m+\frac{\partial Q^m}{\partial R_j^m}\psi_*^m\right)d\Omega$ | (VIII) |
| $V_j^m$ | $\iint_{\Omega_j^m}\left(\frac{\partial G^m}{\partial V_j^m}-C^m\frac{\partial \psi_*^m}{\partial x}+\frac{\partial Q^m}{\partial V_j^m}\psi_*^m\right)d\Omega$ | (IX) |
| $D_{e,x,j}^m$ | $\iint_{\Omega_j^m}\left(\frac{\partial G^m}{\partial D_{e,x,j}^m}+\frac{\partial C^m}{\partial x}\frac{\partial \psi_*^m}{\partial x}+\frac{\partial Q^m}{\partial D_{e,x,j}^m}\psi_*^m\right)d\Omega$ | (X) |
| $D_{e,z,j}^m$ | $\iint_{\Omega_j^m}\left(\frac{\partial G^m}{\partial D_{e,z,j}^m}+\frac{\partial C^m}{\partial z}\frac{\partial \psi_*^m}{\partial z}+\frac{\partial Q^m}{\partial D_{e,z,j}^m}\psi_*^m\right)d\Omega$ | (XI) |
| $C_{0,j}^m$ | $\iint_{\Omega_j^m}\left(\frac{\partial G^m}{\partial C_{0,j}^m}-\frac{D_{e,x}}{L}\frac{\partial \psi_*^m}{\partial x}\Big\vert_{x=0}+\frac{\partial Q^m}{\partial C_{0,j}^m}\psi_*^m\right)d\Omega$ | (XII) |
| $C_{L,j}^m$ | $\iint_{\Omega_j^m}\left(\frac{\partial G^m}{\partial C_{L,j}^m}+\frac{D_{e,x}}{L}\frac{\partial \psi_*^m}{\partial x}\Big\vert_{x=L}+\frac{\partial Q^m}{\partial C_{L,j}^m}\psi_*^m\right)d\Omega$ | (XIII) |
| $b_j$ | $\iint_{\Omega_j^f}\left\{\frac{\partial G^f}{\partial b_j}+\frac{\partial R^f}{\partial b_j}\lambda\phi^f C^f\psi_*^f+\frac{1}{b_j}\left[D_{e,z}\left(\frac{1}{b_j}\frac{\partial C^m}{\partial z}\Big\vert_{z=b_j}-\frac{\partial^2 C^m}{\partial z^2}\Big\vert_{z=b_j}\right)\psi_*^f+G^f\left(b_j,C^f\right)-G^m\left(b_j,C^f\right)\right]+\frac{\partial Q^f}{\partial b_j}\psi_*^f\right\}d\Omega$ | (XIV) |

*Note.* For semi-infinite domain, the integrals (V) and (XII) containing $L$ are zero.

where, $A^f$ and $A^m$ are defined in Equations 26 and 27 below.

Because of the state sensitivities $\psi^f$ and $\psi^m$ are unknowns, we need to set the coefficients in front of $\psi^f$ and $\psi^m$ (i.e., $A^f$ and $A^m$) to be zero in Equation 25. Doing so, we choose the arbitrary functions $\psi_*^f$ and $\psi_*^m$ such that:

$$A^f \equiv \frac{d}{dx}\left(-D_e^f\frac{d\psi_*^f}{dx}\right)-V^f\frac{d\psi_*^f}{dx}+\lambda\phi^f R^f\psi_*^f-\frac{D_{e,z}^m}{b}\frac{\partial\psi_*^m}{\partial z}\Big\vert_{z=b}+\frac{\partial Q^f}{\partial C^f}\psi_*^f+\frac{\partial G^f}{\partial C^f}=0 \tag{26}$$

in $\Omega^f$, and

$$A^m \equiv \frac{\partial}{\partial x}\left(-D_{e,x}\frac{\partial\psi_*^m}{\partial x}\right)+\frac{\partial}{\partial z}\left(-D_{e,z}\frac{\partial\psi_*^m}{\partial z}\right)-V^m\frac{\partial\psi_*^m}{\partial x}+\lambda\phi^m R^m\psi_*^m+\frac{\partial Q^m}{\partial C^m}\psi_*^m+\frac{\partial G^m}{\partial C^m}=0 \tag{27}$$

in $\Omega^m$, with the following boundary conditions:

For the fracture:

$$\psi_*^f(0)=0,\quad 0\le z\le b \tag{28}$$

$$\psi_*^f(L)=0,\quad 0\le z\le b \tag{29}$$

For the matrix:

$$\psi_*^m(0,z)=0,\quad b\le z\le B \tag{30}$$

$$\psi_*^m(L,z)=0,\quad b\le z\le B \tag{31}$$

$$\psi_*^m(x,b)=\psi_*^f(x),\quad 0\le x\le L \tag{32}$$

$$\frac{\partial\psi_*^m}{\partial z}\Big\vert_{z=B}=0,\quad 0\le x\le L \tag{33}$$

The boundary conditions (Equation 28) through (Equation 33) are necessary conditions which are imposed in the course of development of Equation 25 as shown in Appendix A.

Equations 26 and 27 reduce the marginal sensitivity coefficient to:

$$\begin{aligned}\frac{\partial J}{\partial p}=&\iint_{\Omega^f}\left\{\frac{\partial G^f}{\partial p}+\frac{\partial\phi^f}{\partial p}\lambda R^f C^f\psi_*^f+\frac{\partial R^f}{\partial p}\lambda\phi^f C^f\psi_*^f+\frac{\partial D_e^f}{\partial p}\frac{dC^f}{dx}\frac{d\psi_*^f}{dx}\right.\\&-\frac{\partial V^f}{\partial p}C^f\frac{d\psi_*^f}{dx}+\frac{\partial C_L^f}{\partial p}\frac{D_e^f}{L}\frac{d\psi_*^f}{dx}\Big\vert_{x=L}-\frac{\partial C_0^f}{\partial p}\frac{D_e^f}{L}\frac{d\psi_*^f}{dx}\Big\vert_{x=0}\\&+\frac{\partial b}{\partial p}\frac{1}{b}\left[D_{e,z}\left(\frac{1}{b}\frac{\partial C^m}{\partial z}\Big\vert_{z=b}-\frac{\partial^2 C^m}{\partial z^2}\Big\vert_{z=b}\right)\psi_*^f+G^f\left(b,C^f\right)\right.\\&\left.\left.-G^m\left(b,C^f\right)\right]+\frac{\partial Q^f}{\partial p}\psi_*^f\right\}d\Omega\\&\iint_{\Omega^m}\left\{\frac{\partial G^m}{\partial p}+\frac{\partial\phi^m}{\partial p}\lambda R^m C^m\psi_*^m+\frac{\partial R^m}{\partial p}\lambda\phi^m C^m\psi_*^m\right.\\&+\frac{\partial D_{e,x}}{\partial p}\frac{\partial C^m}{\partial x}\frac{\partial\psi_*^m}{\partial x}+\frac{\partial D_{e,z}}{\partial p}\frac{\partial C^m}{\partial z}\frac{\partial\psi_*^m}{\partial z}-\frac{\partial V^m}{\partial p}C^m\frac{\partial\psi_*^m}{\partial x}\\&\left.+\frac{\partial C_L^m}{\partial p}\frac{D_{e,x}}{L}\frac{\partial\psi_*^m}{\partial x}\Big\vert_{x=L}-\frac{\partial C_0^m}{\partial p}\frac{D_{e,x}}{L}\frac{\partial\psi_*^m}{\partial x}\Big\vert_{x=0}+\frac{\partial Q^m}{\partial p}\psi_*^m\right\}d\Omega\end{aligned} \tag{34}$$

Equation 34 determines the marginal sensitivity coefficient for any parameter. If $p_j$ denotes the sensitivity parameter of a subdomain $\Omega_j^f$ of $\Omega^f$ or a subdomain $\Omega_j^m$ of $\Omega^m$, then the global integrals $\iint_{\Omega^f} .d\Omega$ and $\iint_{\Omega^m} .d\Omega$ restrict to the local integrals $\iint_{\Omega_j^f} .d\Omega$ and $\iint_{\Omega_j^m} .d\Omega$, respectively. Note that, for any parameter $p$, $\partial p \,/\, \partial p_j$is zero over the domain $\Omega^f$ (or $\Omega^m$) except the portion $\Omega_j^f$ (or $\Omega_j^m$) in which $\partial p \,/\, \partial p_j$.

Table 1 summarizes the general formulas of the marginal sensitivity coefficients of each parameter. These formulas are deduced from the general expression defined by Equation 34. The table contains 14 formulas which correspond to 14 types of sensitivity parameters that are: the flow and transport parameters $\phi^f$, $R^f$, $V^f$, $D_e^f$, $\phi^m$, $R^m$, $V^m$, $D_{e,x}^m$, and $D_{e,z}^m$, the boundary conditions $C_0^f$, $C_L^f$, $C_0^m$, and $C_L^m$, and the fracture aperture $b$. Therefore, for a given performance measure (i.e., given two functions $G^f$ and $G^m$), the cost of computing any sensitivity coefficient is equivalent to the cost of calculating an integral after solving the forward and adjoint systems.

### 3.2. Discrete Approach

The discrete approach consists in solving first the primary problem (12–19) using a given numerical method (e.g., finite element, finite volume, finite difference). The discretized form of the problem can be written then as follows:

$$[A^f]\{C^{f,m}\} - \{G^f\} = \{0\} \tag{35}$$

$$[A^m]\{C^m\} - \{G^m\} = \{0\} \tag{36}$$

where $[A^f]$ and $[A^m]$ are matrices associated with the discretized numerical scheme of Equations 12 and 13, respectively, $\{G^f\}$ and $\{G^m\}$ are known vectors which depend on the boundary conditions and sink/source terms associated with Equations 12 and 13, and $\{C^{f,m}\}$ and $\{C^m\}$ are the vectors of unknown concentrations at nodes or elements of the discretized numerical grid and associated with Equations 12 and 13, respectively. Note that $\{C^{f,m}\}$ contains nodal values of concentration in both fracture and matrix media, while $\{C^m\}$ contains only nodal values of concentration in porous matrix.

The two linear systems of equations can be combined in a single linear system which combine both the fracture and matrix variables. The system can be written in the following form:

$$[A]\{C\} - \{G\} = \{0\} \tag{37}$$

where the coefficients of matrix $[A]$ contain the nodal values of the parameters of the system (in both fracture and matrix), $\{C\}$ is the unknown vector of nodal concentration values in both fracture and matrix, and $\{G\}$ is a known vector containing the sink/source terms and boundary conditions associated with the fracture and porous matrix media.

The discretized form of the performance measure (Equation 20) may be written as a scalar function of the vector of nodal parameter values $\{p\}$ and the vector of unknown concentrations $\{C\}$: $J = F(\{p\},\{C\})$.

Multiplying (Equation 37) by an arbitrary vector $\{\gamma^*\}$ of the same dimension of $\{C\}$ and summing up to $J$, we get

$$J = F\Big(\{p\},\{C\}\Big) + \{\gamma^*\}^T\Big([A]\{C\} - \{G\}\Big) \tag{38}$$

The vector $\{\gamma^*\}$ is called the discretized adjoint state vector associated with primary system (Equation 37) which must be determined.

Let's consider a given parameter $p_k$ from the vector of nodal parameters $\{p\}$. Differentiating (Equation 38) with respect to $p_k$ yields

$$\frac{\partial J}{\partial p_k} = \frac{\partial F}{\partial p_k} + \frac{\partial F}{\partial \{C\}}\frac{\partial \{C\}}{\partial p_k} + \{\gamma^*\}^T\left(\frac{\partial [A]}{\partial p_k}\{C\} + [A]\frac{\partial \{C\}}{\partial p_k} - \frac{\partial \{G\}}{\partial p_k}\right) \tag{39}$$

Equation 39 can be rearranged to take the following form

$$\frac{\partial J}{\partial p_k} = \frac{\partial F}{\partial p_k} + \{\gamma^*\}^T\left(\frac{\partial [A]}{\partial p_k}\{C\} - \frac{\partial \{G\}}{\partial p_k}\right) + \left(\{\gamma^*\}^T[A] + \frac{\partial F}{\partial \{C\}}\right)\frac{\partial \{C\}}{\partial p_k} \tag{40}$$



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Equation 40 expresses the marginal sensitivity of the performance measure with respect to a given parameter as the sum of three terms. In this equation, there are two unknown vectors that are the adjoint state vector $\{\gamma^*\}$ and the state sensitivity vector $\partial\{C\} / \partial p_k$. The evaluation $\partial\{C\} / \partial p_k$ is computationally expensive. The vector $\{\gamma^*\}$ is however arbitrary (there is no a priori assumption about it). To avoid the computation of $\partial\{C\} / \partial p_k$, we can choose $\{\gamma^*\}$ in such a way that the last term in Equation 40 is zero. In other words, the computation of $\partial\{C\} / \partial p_k$ can be avoided if $\{\gamma^*\}$ is the solution of the following adjoint state system

$$[A]^T\{\gamma^*\} + \frac{\partial F}{\partial\{C\}^T} = \{0\} \tag{41}$$

With Equation 41, the marginal sensitivity coefficient reduces to

$$\frac{\partial J}{\partial p_k} = \frac{\partial F}{\partial p_k} + \{\gamma^*\}^T\left(\frac{\partial[A]}{\partial p_k}\{C\} - \frac{\partial\{G\}}{\partial p_k}\right) \tag{42}$$

The procedure of discrete adjoint approach associated with steady-state transport in fractured media can be then reduced to the resolution of two linear systems: the forward (or primary system) (Equation 37) (to evaluate the concentrations) and the adjoint system (Equation 41) to evaluate the adjoint state vector $\{\gamma^*\}$. The computation of the sensitivity coefficients can be simply obtained from (Equation 42) after using the solutions of the two systems.

It can be observed that the adjoint system (Equation 41) is independent from the number of parameters and it only depends on the performance measure. Therefore, for a given performance measure, the evaluation of sensitivity coefficients requires only the resolution of two linear systems whatever the number of sensitivity parameters. Note that, the matrix of the adjoint system is the transpose of that of the forward system. Therefore, there is no additional computational effort required for the computation of the matrix associated with the adjoint state system.

## 4. Physical Interpretation of The Adjoint States

To provide insight into the physical significance of the adjoint state functions, we assume that the term sources $Q^f$ and $Q^m$ in the fracture and matrix transport operators $L^f$ and $L^m$ are of the forms

$$Q^f = Q^f_* \delta\left(x - x'\right)\delta\left(z - z'\right) \tag{43}$$

$$Q^m = Q^m_* \delta\left(x - x'\right)\delta\left(z - z'\right) \tag{44}$$

where $Q^f_*$ and $Q^m_*$ are the strengths of the sources (associated with the fracture and the rock matrix) introduced in the domain at a given point $\left(x', z'\right)$.

Assume that we are looking for the sensitivities of the performance measure $J$ (as defined by Equation 20) with respect to $Q^f_*$ and $Q^m_*$. Therefore, by applying Equation 34 with $p_k = Q^f_*$ or $p_k = Q^m_*$, we get the following sensitivity coefficients

$$\frac{\partial J}{\partial Q^f_*} = \iint_{\Omega^f}\delta\left(x - x'\right)\delta\left(z - z'\right)\psi^f_* d\Omega = \psi^f_*\left(x'\right) \tag{45}$$

$$\frac{\partial J}{\partial Q^m_*} = \iint_{\Omega^m}\delta\left(x - x'\right)\delta\left(z - z'\right)\psi^m_* d\Omega = \psi^m_*\left(x', z'\right) \tag{46}$$

Equations 45 and 46 indicate that the adjoint state function $\psi^f_*\left(x\right)$ (resp. $\psi^m_*\left(x, z\right)$) associated with the fracture (resp. the rock matrix) is interpreted as the sensitivity coefficient of the performance measure with respect to the steady source strength $Q^f_*$ (resp. $Q^m_*$) at location $x$ in the fracture (resp. $\left(x, z\right)$ in the rock matrix). They represent the rates at which the performance measure varies per unit source of RN $Q^f_*\left(ML^{-1}T^{-1}\right)$ and $Q^m_*\left(ML^{-1}T^{-1}\right)$ at $x$ and $\left(x, z\right)$, respectively. The dimensions of the adjoint states are those of the performance measure $J$ times $\left(LT\ /\ M\right)$, for the steady-state case considered here.

## 5. Exact Solutions in Layered Fractured Medium

In this section, we present exact analytical solutions developed for both primary and adjoint systems in a finite layered system. The exact solutions are obtained by applying a method based on the use of an analytical inversion formula of tridiagonal matrices. These exact solutions are important since they can be considered as reference solutions for the validation of other more complicated numerical solutions. We assume that the geometry of the cross section of the fractured rock formation network corresponds to a series of connected parallel line segments of different thicknesses, and each layer is characterized by its own properties (see Figure 2). The total length $L$ of the domain is partitioned into $N$ layers $l_i$ limited by $x_{i-1}$ and $x_i$ for $i = 1, \ldots, N$, with $x_0 = 0$ and $x_N = L$ (i.e., $l_i = [x_{i-1}, x_i]$ with $x_i$ indicating the position in the $x$ direction of the interface between layer $l_{i-1}$ and layer $l_i$). The length of layer $l_i$ is denoted by $L_i = x_i - x_{i-1}$. It is also assumed that RN transport in the rock matrices is only driven by diffusion in the direction perpendicular to the flow direction in the fracture. In this section, we only present the final solutions. The details of derivation are given in Appendix B.

### 5.1. Exact Solutions of the Forward Problem

Taking into account the above assumptions, the transport equations to be solved for each layer $l_i$ can be written in the following forms (for $x_{i-1} \le x \le x_i$, $b_i \le z \le B$, and $i = 1, \ldots, N$)

$$-D_{e,i}^f \frac{\mathrm{d}^2 C_i^f}{\mathrm{d}x^2} + V_i^f \frac{\mathrm{d}C_i^f}{\mathrm{d}x} + \lambda \phi_i^f R_i^f C_i^f - \frac{D_{e,z,i}^m}{b_i} \frac{\partial C_i^m}{\partial z}\Big|_{z=b_i} + Q_i^f = 0 \tag{47}$$

$$-D_{e,z,i}^m \frac{\partial^2 C_i^m}{\partial z^2} + \lambda \phi_i^m R_i^m C_i^m + Q_i^m = 0 \tag{48}$$

where subscript $i$ indicates the layer number. The exact solutions we are going to present are valid for constant sources (i.e., $Q_i^f$ and $Q_i^m$ are assumed to be independent of the space coordinates).

Equations 47 and 48 are subject to the following boundary conditions:

$$C_1^f(0) = C_0^f, \quad 0 \le z \le b \tag{49}$$

$$C_N^f(L) = C_L^f, \quad 0 \le z \le b \tag{50}$$

$$C_i^m(x, b) = C_i^f(x), \quad 0 \le x \le L \tag{51}$$

$$\frac{\partial C_i^m}{\partial z}\Big|_{z=B} = 0, \quad 0 \le x \le L \tag{52}$$

The exact solutions of system (Equations 47–52) can be written as follows (for $x_{i-1} \le x \le x_i$, $b_i \le z \le B$, and $i = 1, \ldots, N$)

$$C_i^f(x) = (C_i - F_i) e^{\nu_i (x - x_i)} \frac{\sinh\left[\delta_i (x - x_{i-1})\right]}{\sinh(\delta_i L_i)} + (C_{i-1} - F_i) e^{\nu_i (x - x_{i-1})} \frac{\sinh\left[\delta_i (x_i - x)\right]}{\sinh(\delta_i L_i)} + F_i \tag{53}$$

and

$$C_i^m(x, z) = \left[C_i^f(x) + \frac{B_i^m}{\lambda}\right] \frac{\cosh\left[r_i^m (B - z)\right]}{\cosh\left[r_i^m (B - b_i)\right]} - \frac{B_i^m}{\lambda} \tag{54}$$

The details of derivation of Equations 53 and 54 are provided in Appendix B. The definitions of $\nu_i, \delta_i, F_i, B_i^m, r_i^m$ (for $i = 1, \ldots, N$), and $C_i$ (for $i = 1, \ldots, N-1$) can be found (in Appendix B) in Equations B9, B10, B11, B2, B3, and B38, respectively. Note that in Equation 53 we have $C_0 = C_0^f$ and $C_N = C_L^f$.

### 5.2. Exact Solutions for the Adjoint Problem

The adjoint system associated with the forward system 47 and 48 can be defined by the following system of equations (for $x_{i-1} \le x \le x_i$, $b_i \le z \le B$, and $i = 1, \ldots, N$)



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$$-D_{e,i}^{f}\frac{\mathrm{d}^2\psi_{*,i}^{f}}{\mathrm{d}x^2}-V_i^{f}\frac{\mathrm{d}\psi_{*,i}^{f}}{\mathrm{d}x}+\lambda\phi_i^{f}R_i^{f}\psi_{*,i}^{f}-\frac{D_{e,z,i}^{m}}{b_i}\frac{\partial\psi_{*,i}^{f}}{\partial z}\Big|_{z=b_i}+\frac{\partial G^{f}}{\partial C_i^{f}}=0 \tag{55}$$

$$-D_{e,z,i}^{m}\frac{\partial^2\psi_{*,i}^{m}}{\partial z^2}+\lambda\phi_i^{m}R_i^{m}\psi_{*,i}^{m}+\frac{\partial G^{m}}{\partial C_i^{m}}=0 \tag{56}$$

The associated boundary conditions are

$$\psi_{*,1}^{f}(0)=0,\quad 0\le z\le b \tag{57}$$

$$\psi_{*,N}^{f}(L)=0,\quad 0\le z\le b \tag{58}$$

$$\psi_{*,i}^{m}(x,b)=\psi_{*,i}^{f}(x),\quad 0\le x\le L \tag{59}$$

$$\frac{\partial\psi_{*,i}^{m}}{\partial z}\Big|_{z=B}=0,\quad 0\le x\le L \tag{60}$$

The solutions of system (Equations 55–60) depend on the form of functions $G^f$ and $G^m$ defining the performance measure. The solutions presented here are restricted to constant values of $\partial G^f/\partial C_i^f$ and $\partial G^m/\partial C_i^m$. Many performance measures of physical importance can be defined with functions $G^f$ and $G^m$ having constant derivative with respect to $C_i^f$ and $C_i^m$, respectively. This is the case when $G^f$ and $G^m$ are defined as linear combination of $C_i^f$ and $C_i^m$, respectively

$$G^{f}\left(p,C^{f}\right)=\sum_{i=1}^{N}\alpha_i^{f}C_i^{f} \tag{61}$$

$$G^{m}\left(p,C^{m}\right)=\sum_{i=1}^{N}\alpha_i^{m}C_i^{m} \tag{62}$$

where $\alpha_i^f$ and $\alpha_i^m$ are space-independent coefficients which could be dependent of the physical parameters of the system. The values of $\alpha_i^f$ and $\alpha_i^m$ are zeros outside layer $l_i$ but could take nonzero values in $l_i$. The performance measure associated with Equations 61 and 62 can be written as follows

$$J\left(p,C^{f},C^{m}\right)=\sum_{i=1}^{N}\left(\alpha_i^{f}\iint_{\Omega_i^{f}}C_i^{f}d\Omega+\alpha_i^{m}\iint_{\Omega_i^{m}}C_i^{m}d\Omega\right) \tag{63}$$

where $\Omega_i^f=[x_{i-1},x_i]\times[0,b_i]$ and $\Omega_i^m=[x_{i-1},x_i]\times[b_i,B]$ define the regions of the layer $l_i$ in the fracture and the rock matrix, respectively.

The performance measure defined by (Equation 63) is very useful for investigating the sensitivity of the RN mass in a given region. For instance, if we are interested in the RN mass in layer $l_i$ we could set $\alpha_i^f=\phi_i^fR_i^f$ for the mass in the fracture and $\alpha_i^m=\phi_i^mR_i^m$ for the mass in the associated rock matrix. The other values $\alpha_j^f$ (resp. $\alpha_j^m$) for $j\neq i$ should be then set to 0. When $\alpha_i^f=\phi_i^fR_i^f$ and $\alpha_i^m=\phi_i^mR_i^m$ for all $i$, then Equation 63 represents the total RN mass in the system. We could be also interested by the sensitivity of RN mass in the fracture only (i.e., $\alpha_i^f=\phi_i^fR_i^f$ and $\alpha_i^m=0$ for all $i$) or the sensitivity of RN mass in the rock matrix (i.e., $\alpha_i^f=0$ and $\alpha_i^m=\phi_i^mR_i^m$ for all $i$).

If $\partial G^f/\partial C_i^f$ and $\partial G^m/\partial C_i^m$ are constants for all $i$ then the exact solution of the adjoint state system (Equations 55–60) can be deduced from that of the forward problem (47–52) as follows.

From a purely mathematical point of view, the adjoint system (Equations 55–60) is similar to the forward system (Equations 47–52) with the exception that: the sign before the advection term is negative, the term sources are $\partial G^f/\partial C_i^f$ and $\partial G^m/\partial C_i^m$, respectively, and the boundary conditions at $x=0$ and $x=L$ are 0. Therefore, the exact solution of the adjoint system can be deduced from Equations 53 and 54 as follows:

$$\psi_{*,i}^{f}(x)=\left(\psi_{*,i}-F_{*,i}\right)e^{-\nu_i(x-x_i)}\frac{\sinh\left[\delta_i(x-x_{i-1})\right]}{\sinh(\delta_iL_i)}+\left(\psi_{*,i-1}-F_{*,i}\right)e^{-\nu_i(x-x_{i-1})}\frac{\sinh\left[\delta_i(x_i-x)\right]}{\sinh(\delta_iL_i)}+F_{*,i} \tag{64}$$

and

$$\psi_{*,i}^{m}(x,z)=\left[\psi_{*,i}^{f}(x)+\frac{B_i^{m}}{\lambda}\right]\frac{\cosh\left[r_i^{m}(B-z)\right]}{\cosh\left[r_i^{m}(B-b_i)\right]}-\frac{B_{*,i}^{m}}{\lambda} \tag{65}$$



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**Table 2**
*Input Parameters Used for the Verification of Analytical Results*

| N° | $L$ (m) | $b$ (m) | $u$ (m/yr) | $D_e^f$ (m²/yr) | $\phi^m$ (−) | $\rho$ (kg/m³) | $D_{e,z}^m$ (m²/yr) | $K^f$ (m) | | $K^m$ (m³/kg) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Case 1 | Case 2 | Case 1 | Case 2 |
| 1 | 50.0 | $5.0 \times 10^{-3}$ | 10.0 | 1.0 | 0.01 | 2000.0 | $1.0 \times 10^{-4}$ | $5.0 \times 10^{-3}$ | $1.5 \times 10^{-2}$ | $5.0 \times 10^{-4}$ | $1.5 \times 10^{-3}$ |
| 2 | 75.0 | $4.0 \times 10^{-3}$ | 12.5 | 1.0 | 0.008 | 2300.0 | $1.6 \times 10^{-4}$ | $8.0 \times 10^{-3}$ | $8.0 \times 10^{-3}$ | $6.978 \times 10^{-4}$ | $1.2 \times 10^{-3}$ |
| 3 | 100.0 | $3.0 \times 10^{-3}$ | 16.666 | 1.0 | 0.006 | 2600.0 | $3.6 \times 10^{-4}$ | $2.7 \times 10^{-2}$ | $5.4 \times 10^{-2}$ | $1.158 \times 10^{-3}$ | $1.25 \times 10^{-3}$ |
| 4 | 150.0 | $2.0 \times 10^{-3}$ | 25.0 | 1.0 | 0.004 | 2650.0 | $2.0 \times 10^{-4}$ | $1.0 \times 10^{-2}$ | $1.0 \times 10^{-2}$ | $1.059 \times 10^{-3}$ | $7.5 \times 10^{-4}$ |
| 5 | 200.0 | $1.5 \times 10^{-3}$ | 33.333 | 1.0 | 0.002 | 2700.0 | $6.0 \times 10^{-5}$ | $3.0 \times 10^{-3}$ | $4.5 \times 10^{-3}$ | $7.41 \times 10^{-4}$ | $2.0 \times 10^{-3}$ |

*Note*. Case 1 (resp. case 2) corresponds to the case of long half-life RN ($^{237}$Np) with $T_{1/2} = 2.3 \times 10^6$ yr resp. short half-life RN ($^{245}$Cm) with $T_{1/2} = 8.5 \times 10^3$ yr, where $T_{1/2}$ is the RN half-life.

with

$$B_{*,i}^m = \frac{\partial G^m / \partial C_i^m}{\phi_i^m R_i^m} \tag{66}$$

$$F_{*,i} = -\frac{\partial G^f / \partial C_i^f}{r_{ai}} - \frac{c_{fi} B_{*,i}^m}{\lambda^{1/2} r_{ai}} \tag{67}$$

where $\psi_{*,i}$ is obtained from Equation B38 by replacing $V_i^f$ by $-V_i^f$ in all the mathematical expressions leading to Equation B38. In addition, in Equation 64 we have $\psi_{*,0} = \psi_{*,N} = 0$.

## 6. Numerical Experiments

The adjoint sensitivity model presented in this work is illustrated with various numerical experiments. The numerical examples are split into two distinct parts. The purpose of the first part is to provide some insights from the adjoint states and to validate, through developed numerical solutions, the exact analytical model which is based on the continuous adjoint approach. In the second part, the discrete adjoint sensitivity approach is used and applied to a real field case represented by a site currently being considered for underground nuclear storage in Northern Switzerland, "Zürich Nordost". Two types of performance measures are considered: RN masses which are used in the validation part and RN flux which is used in the field case application.

### 6.1. Verification of Analytical Results

The test cases used to validate analytical results refer to transport of a RN in a layered geologic formation composed of five horizontal parallel layers where the fracture is normal to the rock layers, as shown in Figure 2 with piecewise constant parameters. Transport in the rock matrices is assumed to be in two directions (parallel and normal to the fracture) but in a first stage (where we compare numerical and analytical results) only diffusion in the direction normal to the fracture is taken into account. Two cases are considered: Case 1 which refer to $^{237}$Np (i.e., long half-life) and case 2 referring to $^{245}$Cm (short half-life). The retardation factors in the $i^{th}$ layer of the fracture $R_i^f$ and the rock matrix $R_i^m$, respectively, are given by (Neretnieks et al., 1982)

$$R_i^f = 1 + \frac{K_i^f}{b_i} \tag{68}$$

and

$$R_i^m = 1 + \frac{1 - \phi_i^m}{\phi_i^m} \rho_i^m K_i^m \tag{69}$$

where, $\rho_i^m$ is the bulk density of the rock matrix $\left(ML^{-3}\right)$, $K_i^f$ is the surface distribution coefficient in the fracture $(L)$, and $K_i^m$ is the distribution coefficient in the rock matrix $(L^3M^{-1})$.

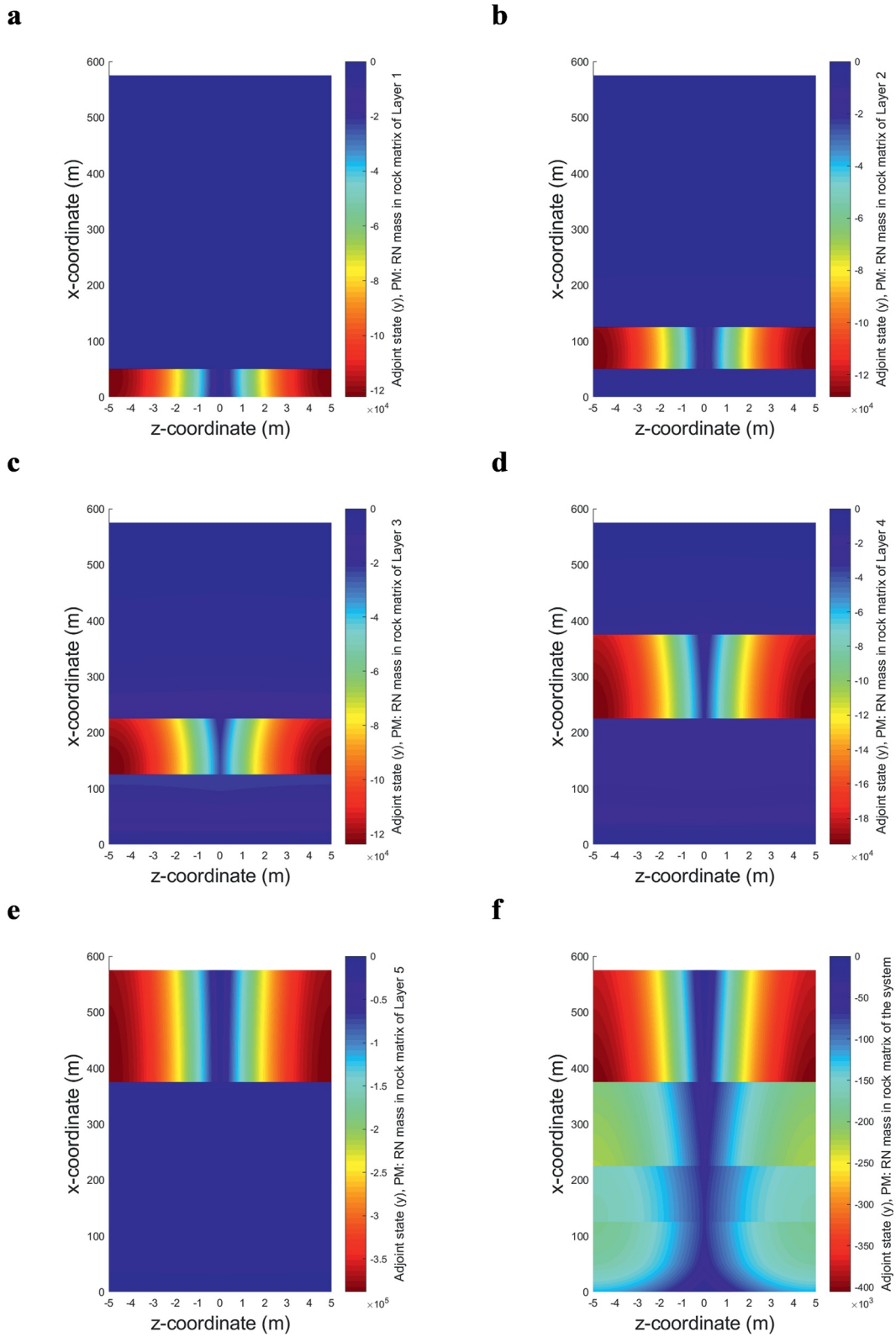



**Figure 3.** Case 1 (long half-life radionuclide (RN) $^{237}$Np). Spatial distribution of the adjoint state function associated with several performance measures representing: (a) RN mass in rock matrix of layer 1, (b) RN mass in rock matrix of layer 2, (c) RN mass in rock matrix of layer 3, (d) RN mass in rock matrix of layer 4, (e) RN mass in rock matrix of layer 5, and (f) RN mass in rock matrix of the system, respectively.



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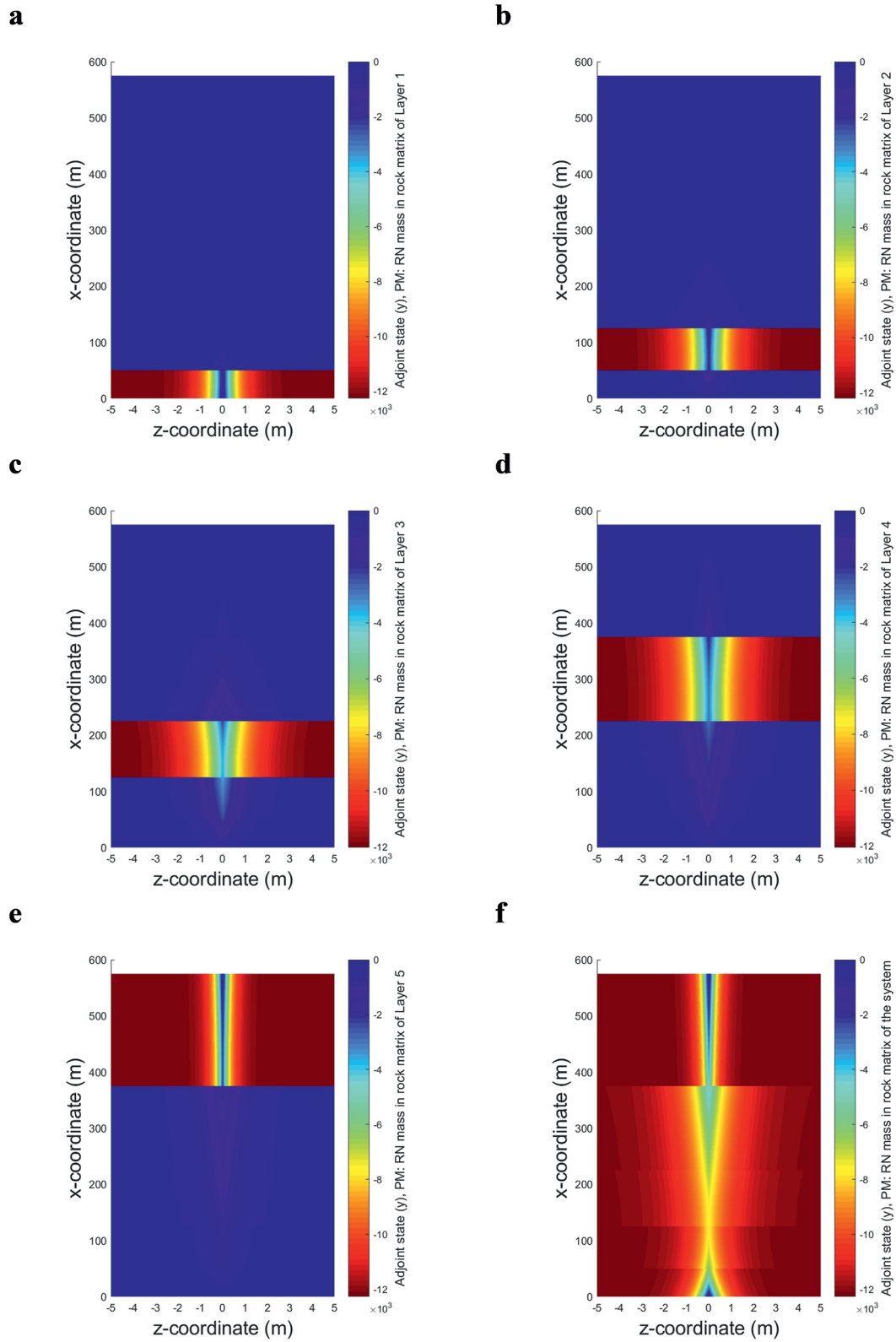



**Figure 4.** Case 2 (short half-life radionuclide (RN) $^{245}$Cm). Spatial distribution of the adjoint state function associated with several performance measures representing: (a) RN mass in rock matrix of layer 1, (b) RN mass in rock matrix of layer 2, (c) RN mass in rock matrix of layer 3, (d) RN mass in rock matrix of layer 4, (e) RN mass in rock matrix of layer 5, and (f) RN mass in rock matrix of the system, respectively.



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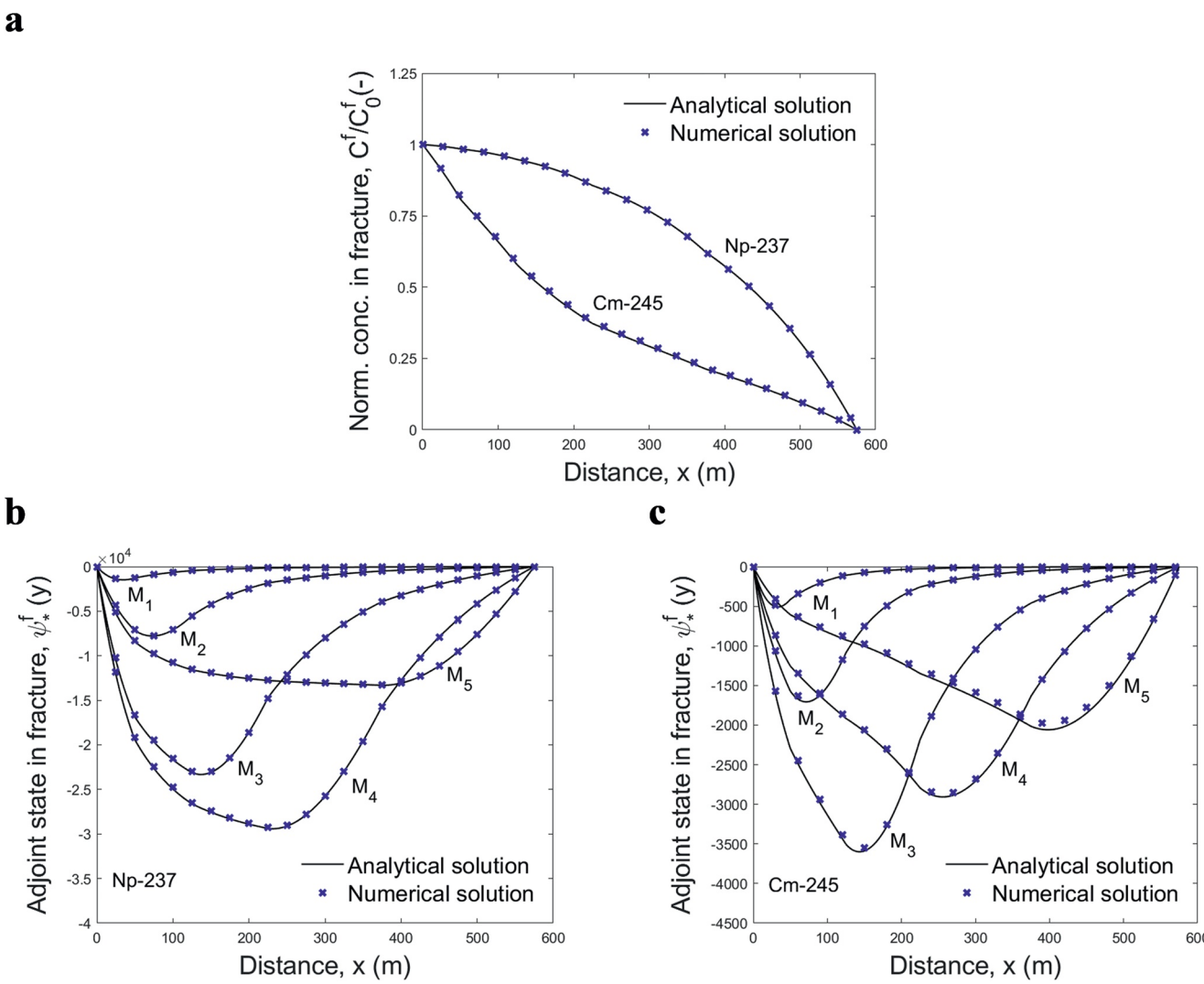



**Figure 5.** Comparison between analytical and numerical results. (a) radionuclide (RN) concentrations for half-life RN $^{237}$Np (bottom left) and short half-life RN $^{245}$Cm. (b) adjoint state functions associated with RN mass $M_i$ in layer $l_i$ for long half-life RN $^{237}$Np, and (c) short half-life RN $^{245}$Cm (right), respectively.

The input parameters used in this test case are taken from Gureghian et al. (1993) and are also presented in Table 2. Note that layer 5 in Gureghian et al. (1993) was semi-infinite while in the present study all layers have finite lengths. The fracture is assumed to be an open channel with porosity equal to one in all layers. The values of boundary conditions as defined in Equations 14–17 are as follow: $C_0^f > 0$, and $C_L^f = C_0^m = C_L^m = 0$.

The performance measures are the RN masses in the individual porous matrix layers $M_i\,(i = 1, \cdots, 5)$ and the total RN mass in the porous matrix $M = \sum_{i=1}^{5} M_i$.

Figures 3 and 4 show the spatial distribution of the analytical adjoint states obtained for each performance measure and associated with case 1 and case 2, respectively. The spatial distribution of the adjoint states is symmetric with respect to the vertical axis passing through the fracture center. For each performance measure defined as the RN mass $M_i$ in the rock matrix of layer $l_i$, the higher values of the adjoint state (i.e., in absolute values) take place in layer $l_i$.

By definition (as shown in Section 4), values for the adjoint state functions at a given location in the rock matrix provide the change in the value of the performance measure given a unit sink/source of RN. The adjoint state functions are negative over the entire domain indicating that an influx of RN at a given loca-



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tion of the rock matrix would correspond to a sink, since a RN sink reduces the mass in the rock matrix. Therefore, if the performance measure is defined as RN mass, the adjoint state function at a given location is defined as the sensitivity coefficient of the mass with respect to the steady sink at this location.

Inspections of Figures 3 and 4 indicate that for each performance measure, the adjoint state functions have the same spatial distribution for both cases. Note that when the performance measure is defined as the total mass of RN in the whole rock matrix system, the high values of adjoint sates (in absolute values) cover the entire rock matrix of the domain. The difference in the values between cases 1 and 2 is mainly due to the difference in physical properties of both RNs (differences in sorption coefficients as shown in Table 2).

To validate the developed analytical adjoint state functions, the system of transport Equations 12–19 and the system of adjoint state Equations 26–33 are solved numerically, and the obtained numerical solutions are compared with the analytical ones. Both systems of equations are discretized and solved numerically in MATLAB using the finite volume scheme. Figure 5 depicts the comparison between analytical and numerical results along the vertical axis of the fracture. Figure 5a shows the concentration profiles associated with both RNs ($^{237}$Np and $^{245}$Cm) while the comparisons between the adjoint state functions associated with both RNs are shown on Figures 5b and 5c, respectively. The figures show excellent agreements between analytical and numerical results. For each layer $l_i$, the adjoint state profile associated with the performance measure $M_i$ has a minimum located inside the layer $l_i$. This indicates that the position of the point at which an influx of RN has the greatest impact on the RN mass $M_i$ is located inside the layer $l_i$.

Sensitivity coefficients $\partial J / \partial p$ are calculated based on the formulas presented in Table 1. Four types of parameters are considered which are the effective diffusion coefficients of the rock matrices, the retardation factors of the rock matrices, the Darcy velocities in the fractures, and the retardation factors of the fractures. The first three parameters would have the main effects on the RN masses in the porous rock matrices as they represent the physical properties of the rock matrices (i.e., the effective diffusion coefficients which drive the exchange between the fracture and rock matrix, and the sorption coefficients which quantify the mass of RN in the matrix) and the fluid flow property (represented by the Darcy velocity of the fracture) which drives the RN transport mechanism through the fracture. The retardation factors of the fractures would be expected to have a smaller impact on the RN mass in the rock matrices.

To verify the analytically derived sensitivity coefficients, the sensitivities are also calculated using a numerical approach based on the forward scheme of the FDA which is given by

$$\frac{\partial J}{\partial p} \approx \frac{J\left(p+h\right)-J\left(p\right)}{h} \tag{70}$$

where $h$ is the step size. A previous study (Bard, 1974) shows that the step size should be small enough to reduce the truncation error and large enough to cause a reasonable change in the significant figures of parameter $p$. A judicious choice of the step size would be $h = \varepsilon p$, where $\varepsilon$ is a small parameter. Preliminary numerical experiments show that $\varepsilon = 10^{-6}$ is a convenient choice for the type of problems investigated in the present study.

Table 3 presents the sensitivity coefficients associated with cases 1 and 2, respectively, and obtained from both the analytical adjoint state approach and the numerical FDA approach. It can be observed that analytical and numerical sensitivities are in excellent agreements for all parameters. The results show that the sensitivity coefficients corresponding to the effective diffusion coefficients of the rock matrices are several orders of magnitude lager than the other sensitivities. For both RNs, the highest sensitivity coefficient is the one corresponding to the effective diffusion coefficient of the rock matrix of layer 5. However, these results should be taken with caution. Indeed, the sensitivities of all types of parameters are qualitatively not comparable as they have different units. To overcome this problem, it is recommended to use instead the normalized sensitivity coefficient which is defined as $\left(p / J\right) \cdot \partial J / \partial p$. It can be thought of as the percent change in the performance measure $J$ caused by 1% change in the value of parameter $p$ (Wilson & Metcalfe, 1985). By using the normalized sensitivities, all sensitivity coefficients become dimensionless and the comparison between them becomes more judicious. The normalized sensitivity coefficients associated with each parameter are also shown in Table 3 for both cases. The results show that the most sensitive parameter in case 1 is the retardation factor of layer 4 followed by the Darcy velocity of layer 1, while in case 2, the Darcy velocity is the most sensitive parameter followed by both the retardation factor and the effective diffusion coefficient

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**Table 3**

*Analytical (Adjoint State Approach) and Numerical (FDA Approach) Sensitivity Coefficients $\partial J/\partial p$ Associated With Case 1 ($^{237}$Np) for the Performance Measure Representing the Total Mass in the Rock Matrix of the Fractured System*

| | | Parameters | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $D^m_{e,z}$ | | | $R^m$ | | | $R^f$ | | | $V^f$ | | |
| | | Sensitivity coefficients | | Normalized sens. coef. | Sensitivity coefficients | | Normalized sens. coef. | Sensitivity coefficients | | Normalized sens. coef. | Sensitivity coefficients | | Normalized sens. coef. |
| | N° | Adjoint-state | FDA | Adjoint-state | Adjoint-state | FDA | Adjoint-state | Adjoint-state | FDA | Adjoint-state | Adjoint-state | FDA | Adjoint-state |
| Case 1 ($^{237}$Np) | 1 | $0.5806 \times 10^5$ | $0.5806 \times 10^5$ | 0.0013 | 2.3402 | 2.3402 | 0.0538 | −0.0023 | −0.0023 | $-0.0108 \times 10^{-4}$ | 137.3345 | 137.3345 | 0.3158 |
| | 2 | $0.8426 \times 10^5$ | $0.8426 \times 10^5$ | 0.0031 | 2.7137 | 2.7137 | 0.1249 | −0.0055 | −0.0055 | $-0.0376 \times 10^{-4}$ | 33.9346 | 33.9346 | 0.0975 |
| | 3 | $0.7376 \times 10^5$ | $0.7376 \times 10^5$ | 0.0061 | 2.5643 | 2.5643 | 0.2947 | −0.0053 | −0.0053 | $-0.1224 \times 10^{-4}$ | −5.2171 | −5.2171 | −0.0200 |
| | 4 | $2.5344 \times 10^5$ | $2.5344 \times 10^5$ | 0.0117 | 2.0919 | 2.0919 | 0.3366 | −0.0033 | −0.0033 | $-0.0454 \times 10^{-4}$ | −20.7942 | −20.7942 | −0.1195 |
| | 5 | $8.2730 \times 10^5$ | $8.2730 \times 10^5$ | 0.0114 | 0.6147 | 0.6147 | 0.1413 | −0.0007 | −0.0007 | $-0.0049 \times 10^{-4}$ | −33.7517 | −33.7517 | −0.1294 |
| Case 2 ($^{245}$Cm) | 1 | $-0.1195 \times 10^5$ | $-0.1196 \times 10^5$ | −0.0256 | −0.0040 | −0.0040 | −0.0256 | −0.0063 | −0.0063 | −0.0005 | 2.1419 | 2.1419 | 0.4579 |
| | 2 | $-0.2361 \times 10^5$ | $-0.2364 \times 10^5$ | −0.0809 | −0.0110 | −0.0110 | −0.0809 | −0.0130 | −0.0130 | −0.0008 | 1.0976 | 1.0976 | 0.2933 |
| | 3 | $-0.2195 \times 10^5$ | $-0.2197 \times 10^5$ | −0.1691 | −0.0148 | −0.0148 | −0.1702 | −0.0126 | −0.0126 | −0.0051 | 0.8865 | 0.8865 | 0.3159 |
| | 4 | $-0.3213 \times 10^5$ | $-0.3217 \times 10^5$ | −0.1375 | −0.0130 | −0.0130 | −0.1382 | −0.0117 | −0.0117 | −0.0015 | 0.6702 | 0.6702 | 0.3582 |
| | 5 | $3.3474 \times 10^5$ | $3.3474 \times 10^5$ | 0.4294 | 0.0075 | 0.0075 | 0.4294 | −0.0050 | −0.0050 | −0.0004 | −0.6283 | −0.6283 | −0.2239 |



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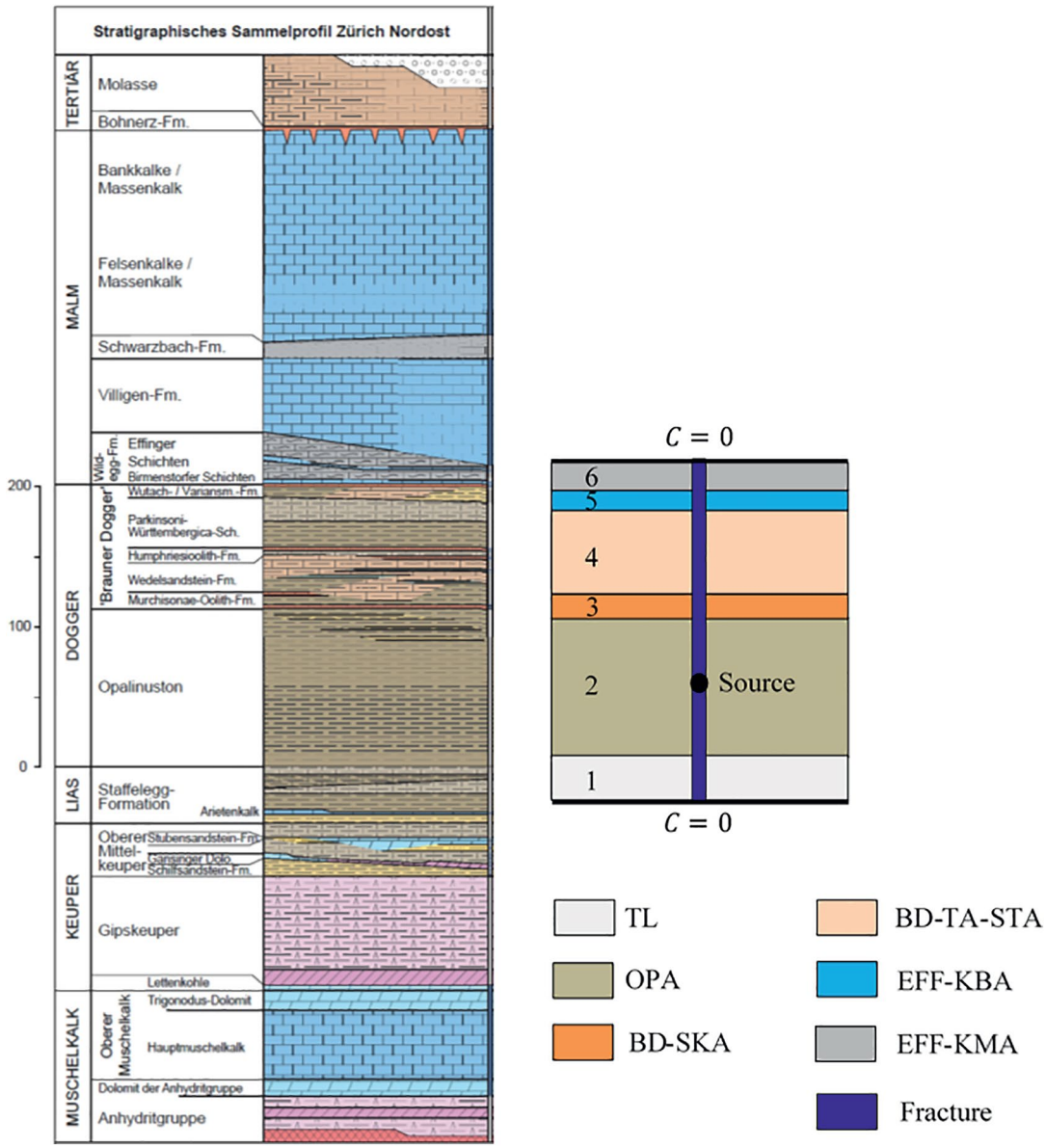



**Figure 6.** (left) Schematic stratigraphic-hydrogeological profile for the siting area Zürich Nordost (reproduced with permission of Nagra, NTB 14-03, Nagra, 2014, Figures A4-3). (right) Modeling concept for radionuclide transport model including vertical fracture.

of layer 5. These results indicate that the results of the sensitivity analysis would depend on the properties of RN. For instance, $^{237}$Np and $^{245}$Cm have different sorption coefficients (and so different retardation factors). The results for case 2 show that despite the large difference in values of sensitivities associated with the effective diffusion coefficients and retardation factors (5–6 orders of magnitude difference), the normalized sensitivities of both type of parameters are almost identical and so they represent the same importance.

### 6.2. Field Case Application

The purpose of this example is to validate and apply the discrete adjoint approach to a real field case represented by a site currently being considered for underground nuclear storage in Northern Switzerland, "Zürich Nordost." In this application, diffusion in the fracture in direction parallel to the flow direction is considered. Such problem has no analytical solution, so numerical solutions are used to compute the sensitivity coefficients using the discrete adjoint approach. The Zürich Nordost model is described in detail

**Table 4**
*Sensitivity Coefficients Obtained by the Adjoint State Method (Discrete Approach) and the FDA Approach for the Example Presented in Section 6.2 Using the Reference Values of All Parameters*

| | Parameters | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $D^m_{e,x}$ | | | $D^m_{e,z}$ | | | $K^m$ | | | $T$ | | |
| | Sensitivity coefficients | | | Sensitivity coefficients | | | Sensitivity coefficients | | | Sensitivity coefficients | | |
| Layer | N° | Adjoint-state | FDA | N° | Adjoint-state | FDA | N° | Adjoint-state | FDA | N° | Adjoint-state | FDA |
| TL | 1 | $-3.4356 \times 10^{-7}$ | $-3.4356 \times 10^{-7}$ | 7 | $-1.8187 \times 10^{-7}$ | $-1.8186 \times 10^{-7}$ | 13 | $-0.0266 \times 10^{-8}$ | $-0.0266 \times 10^{-8}$ | 19 | $2.5248 \times 10^{-7}$ | $2.5248 \times 10^{-7}$ |
| OPA | 2 | $-5.4599 \times 10^{-8}$ | $-5.4596 \times 10^{-8}$ | 8 | $-2.4171 \times 10^{-6}$ | $-2.4169 \times 10^{-6}$ | 14 | $-0.6373 \times 10^{-8}$ | $-0.6373 \times 10^{-8}$ | | | |
| BD-SKA | 3 | $1.4669 \times 10^{-7}$ | $1.4668 \times 10^{-7}$ | 9 | $-1.7836 \times 10^{-7}$ | $-1.7836 \times 10^{-7}$ | 15 | $-0.0956 \times 10^{-8}$ | $-0.0956 \times 10^{-8}$ | | | |
| BD-TA-STA | 4 | $-2.2399 \times 10^{-7}$ | $-2.2398 \times 10^{-7}$ | 10 | $-2.6637 \times 10^{-6}$ | $-2.6635 \times 10^{-6}$ | 16 | $-0.6540 \times 10^{-8}$ | $-0.6540 \times 10^{-8}$ | | | |
| EFF-KBA | 5 | $-2.7634 \times 10^{-6}$ | $-2.7632 \times 10^{-6}$ | 11 | $-1.4975 \times 10^{-6}$ | $-1.4975 \times 10^{-6}$ | 17 | $-0.0061 \times 10^{-8}$ | $-0.0061 \times 10^{-8}$ | | | |
| EFF-KMA | 6 | $3.6964 \times 10^{-7}$ | $3.6964 \times 10^{-7}$ | 12 | $-1.9860 \times 10^{-6}$ | $-1.9860 \times 10^{-6}$ | 18 | $-0.0039 \times 10^{-8}$ | $-0.0039 \times 10^{-8}$ | | | |

in Hayek et al. (2019, 2020). In brief, the model consists of six geological layered formations denoted from bottom to top by TL (Toniger Lias formation), OPA (Opalinus clay host rock), BD-SKA (Sandkalkabfolgen formation), BD-TA-STA (Tonige Abfolgen and Sandig-to-nige Abfolgen formations), EFF-KBA (Kalkbankabfolgen formation), and EFF-KMA (Kalkmergelabfolgen formation), respectively. Each geological layer is characterized by its proper physical properties (i.e., porosity, diffusion coefficient, and sorption coefficient). The model extends to a lateral distance of 20 m. It is assumed that a vertical high permeable fracture with constant thickness equal to 10 cm is crossing the layered model from bottom to top in the middle so that it crosses the repository center. This is a critical situation which can be caused by rock cracking at the repository level due to an overpressure resulting from thermo-hydraulic effects.

### 6.2.1. Base Case Results

We consider the two-dimensional layered system described above which contains a vertical fracture. A steady source of RN that is Uranium-235 ($^{235}$U) is assumed to be released in the fracture at the center of host rock OPA with a constant rate equal to $2 \times 10^{-6}$ m$^3$/year. Zero concentration boundary conditions are imposed at the top and bottom boundaries of the layered system (see Figure 6, right). The system of Equations 12–19 is discretized numerically using a centered finite volume scheme. Transport in the porous matrix of the layered system is assumed to be dominated by diffusion so that $V^m$ in Equation 13 is set to zero. The fracture transmissivity value used in the numerical simulation is $5 \times 10^{-12}$ m$^2$/s and the vertical hydraulic gradient (from bottom to top of the model) is assumed to be equal to 1. Bearing in mind that the fracture thickness is equal to 10 cm, we get a vertical Darcy velocity in the fracture equal to $5 \times 10^{-11}$ m/s. The layer thicknesses and the remaining transport parameters (mainly porosity, effective diffusion coefficient, and sorption coefficient associated with each geological layer) are taken from Table 4 (reference values) of Hayek et al. (2019).

The performance measure of interest is the total diffusive flux leaving the system at the top of the model (i.e., the sum of RN fluxes leaving the fracture and the porous matrix at the top). For the discrete approach used in this application, the general expression of the performance measure may be written as $J = \sum_{i \in I_f} \left( -D_i^f \partial C_i^f \,/\, \partial x \right) + \sum_{i \in I_m} \left( -D_i^m \partial C_i^m \,/\, \partial x \right)$, where $I_f$ and $I_m$ are the parts of the top interface including in the fracture and the rock matrix, $C_i^f$ and $C_i^m$ are the RN concentrations at $I_f$ and $I_m$, and $D_i^f$ and $D_i^m$ are the effective diffusion coefficients of $I_f$ and $I_m$, respectively. The transport parameters which are assumed to be sensitive parameters are the transversal and lateral effective diffusive coefficients of each porous matrix ($D^m_{e,x}$ and $D^m_{e,z}$), the distribution coefficient of each geological layer ($K^m$), as well as the fracture transmissivity ($T$). The total number of sensitive parameters is then 19.

Figure 7 shows the spatial distribution and contour lines of RN concentration (left) and adjoint state (right) associated with $^{235}$U release. These results are obtained by solving the linear systems 37 and 41, respectively. The symbols "<" at the right of each plot indicate the interface location between two consecutive layers. Figure 7 (left) shows how the RN spreads around the source location and how the contour lines converge toward the high permeable fracture. Contour plots of the adjoint state sensitivity function $\gamma^*$ are shown in Figure 7 (right) for the total diffusive flux performance measure. The calculated diffusive flux through the interface defining the top of the model is $5.3751 \times 10^{-8}$ kg/(m s). Values of adjoint state function are defined at the centers of elements of the numerical grid and provide change in the value of the performance measure given a unit volume influx of RN at the element. It can be observed that the adjoint state function is positive over the entire grid indicating that an influx of RN at any point in the system will result in an increase in RN flux. Largest values of the adjoint state function (i.e., the region of largest effects to the RN flux) are in the upper layer (EFF-KMA) in the vicinity of the fracture and close to the interface. This is reasonable because an influx of RN in that region will result in large concentration gradient and so in a larger RN flux.



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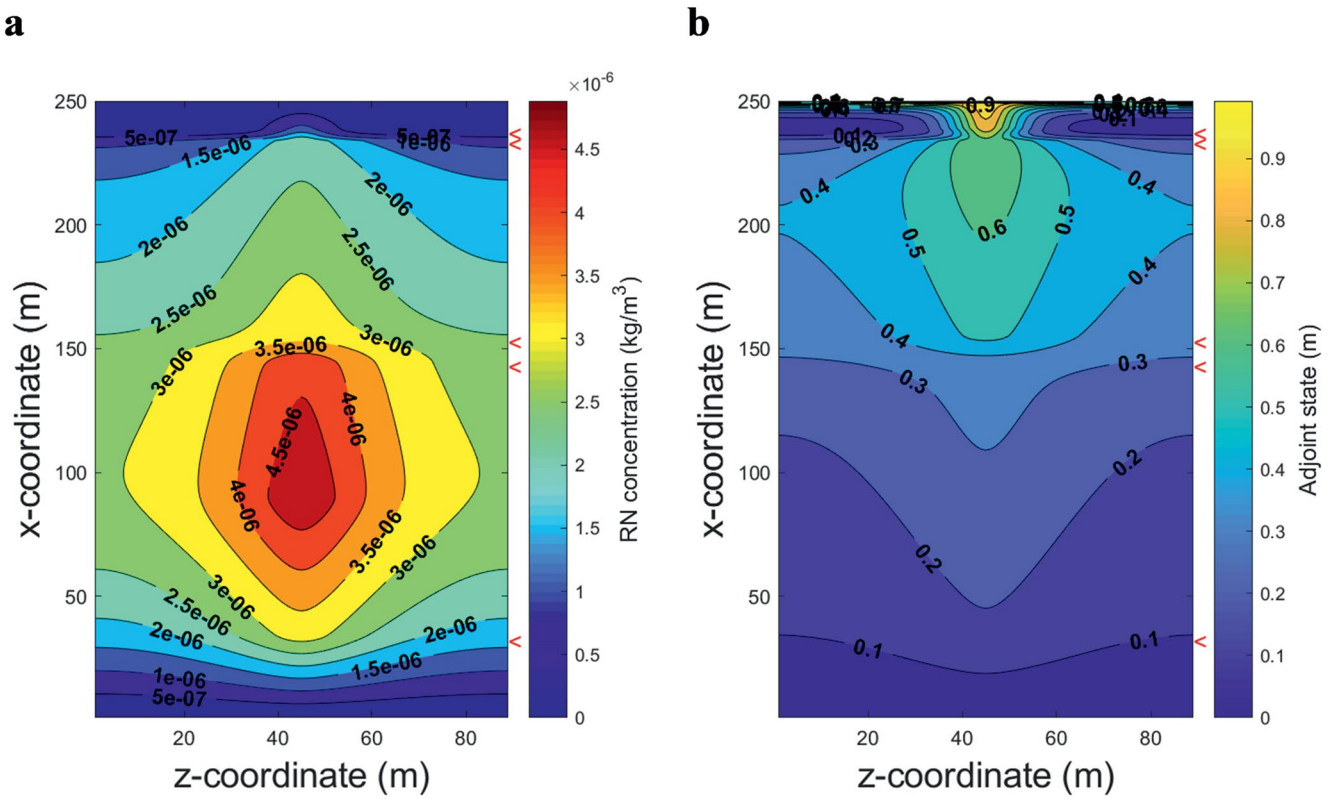



**Figure 7.** Spatial distributions and contour plots of radionuclide concentration (left) and adjoint state (right) associated with $^{235}U$ release.

Table 4 shows for the 19 studied parameters the sensitivity coefficients obtained from the discrete adjoint approach and the finite difference approach, respectively. The adjoint state sensitivities are obtained from Equation 42 after solving the discrete forward and adjoint systems 37 and 41, respectively. The table shows excellent agreements between sensitivity values obtained from both approaches. The main difference is in the computational effort required for each approach to get the sensitivity coefficients. Using FDA, the sensitivities were obtained after solving Equation 37 20 times, while the sensitivity coefficients obtained from the adjoint state approach are obtained after solving only two systems (i.e., system 37 and 41). The results in Table 4 indicate that the effective diffusion coefficient (in vertical direction) of layer EFF-KBA is the most important parameter.

The normalized sensitivity coefficients associated with four types of parameters (lateral and transversal effective diffusion coefficients, retardation factor, and fracture transmissivity) are shown in Figure 8. Panels (a–c) of Figure 8 display the 2-D spatial distribution and contour lines associated with $D_{e,x}$, $D_{e,z}$, and $R$, respectively, while part d depicts the 1-D normalized sensitivity profile (along the vertical fracture) associated with the fracture transmissivity $T$. These normalized sensitivity coefficients represent the percent change of the performance measure (i.e., the RN flux) for a 1% increase of the parameter value. A negative value indicates that a parameter increase (resp. decrease) will result in a decrease (resp. increase) of the performance measure. It should be noted that this kind of mapping of the sensitivity coefficients over the entire domain (as shown in parts a, b, and c of Figure 8) is specific to the adjoint state method and it is in practice not possible to obtain with other methods (e.g., sensitivity method or FDA method) due to their computational costs (i.e., the number of model evaluations is equal to the number of grid elements for these methods while only one adjoint system is required with the adjoint method).

Inspection of Figure 8 indicates that the RN flux at the top of the model is most sensitive to fracture transmissivity values of elements located in the fracture at the levels of layers OPA and BD-TA-STA, respectively, and to the vertical effective diffusion coefficient values of the fracture elements located in the region situated between the center of the host rock OPA to the top of the model. In these fracture elements the



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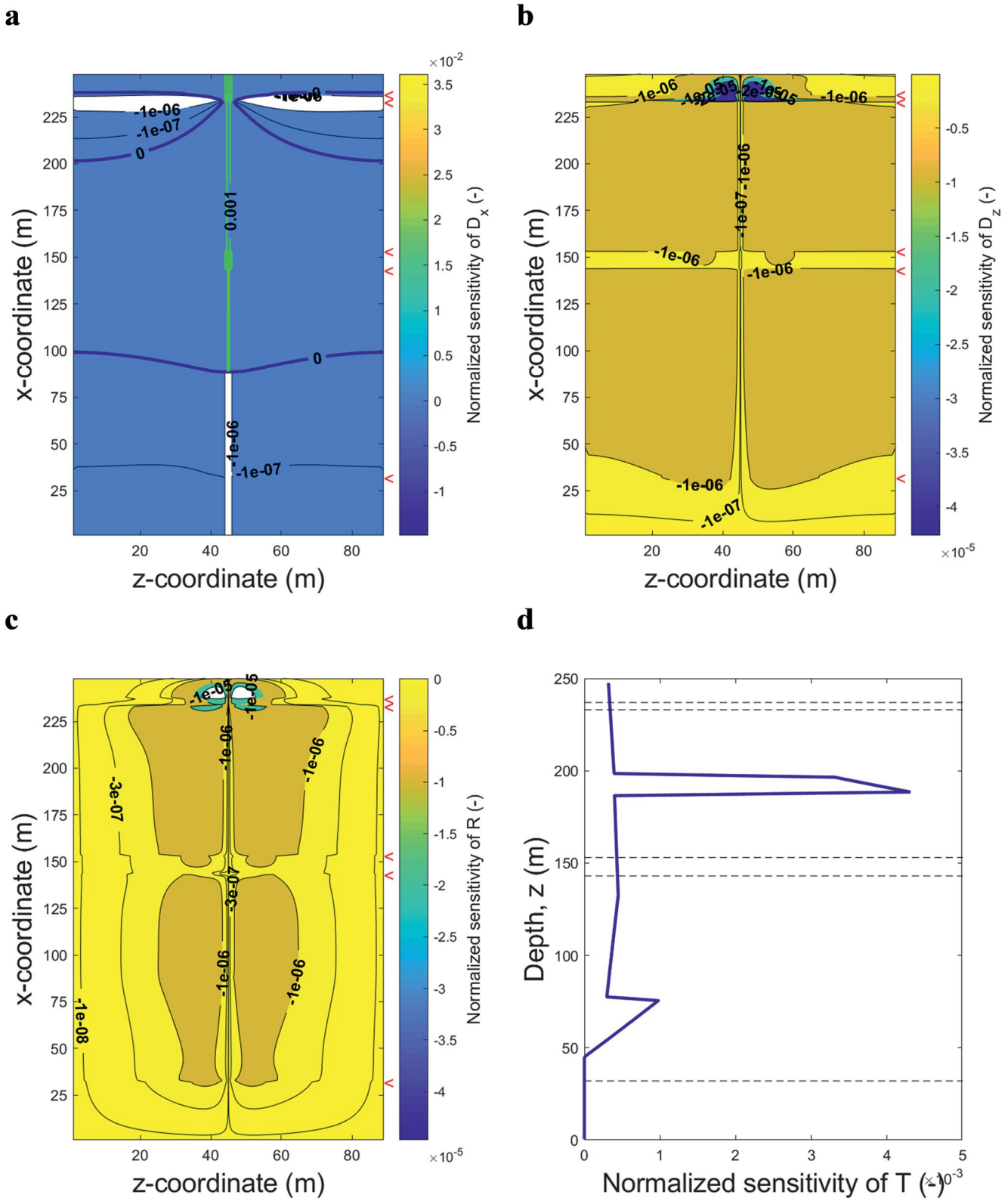



**Figure 8.** Contour plot of normalized adjoint state sensitivity for radionuclide flux performance measure at top of Zürich Nordost model: (a) sensitivity associated with vertical effective diffusion coefficient, (b) sensitivity associated with horizontal effective diffusion coefficient, and (c) sensitivity associated with retardation factor. (d) Vertical distribution of normalized adjoint sensitivity with respect to fracture transmissivity.



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normalized sensitivity coefficients are positive and are generally larger than those for the other zones. This quantifies the importance of the RN flux released from the fracture that results from the higher values of transmissivity and effective diffusion coefficient in those elements.

### 6.2.2. Differential Sensitivity Analysis

In this section, the developed discrete adjoint approach is coupled with the differential sensitivity analysis (DSA) method (Helton, 1993) in order to rank the parameters with respect to their contributions to the uncertainty of the performance measure (i.e., the RN flux at the top of the model).

The DSA method is based on using a Taylor series to approximate the model around a selected base case of the parameter vector. The base case parameter set used here corresponds to the reference values used above (i.e., the set of reference parameters used in Table 4 of Hayek et al., 2019). The list of 19 parameters used in this example are presented in Table 4. We shall denote the vector of base parameters (i.e., $p_1 = D_{e,x}^{\mathrm{TL}}$, $p_2 = D_{e,x}^{\mathrm{OPA}}$, $p_3 = D_{e,x}^{\mathrm{BD-SKA}}$, $p_4 = D_{e,x}^{\mathrm{BD-TA-STA}}$, $p_5 = D_{e,x}^{\mathrm{EFF-KBA}}$, $p_6 = D_{e,x}^{\mathrm{EFF-KMA}}$, $p_7 = D_{e,z}^{\mathrm{TL}}$, $p_8 = D_{e,z}^{\mathrm{OPA}}$, $p_9 = D_{e,z}^{\mathrm{BD-SKA}}$, $p_{10} = D_{e,z}^{\mathrm{BD-TA-STA}}$, $p_{11} = D_{e,z}^{\mathrm{EFF-KBA}}$, $p_{12} = D_{e,z}^{\mathrm{EFF-KMA}}$, $p_{13} = K^{m,\mathrm{TL}}$, $p_{14} = K^{m,\mathrm{OPA}}$, $p_{15} = K^{m,\mathrm{BD-SKA}}$, $p_{16} = K^{m,\mathrm{BD-TA-STA}}$, $p_{17} = K^{m,\mathrm{EFF-KBA}}$, $p_{18} = K^{m,\mathrm{EFF-KMA}}$, and $p_{19} = T$) by $\mathbf{p}_0$ and the RN flux at the top of the model by $f$. A first order Taylor series approximation of the RN flux around the base value has the form

$$f(\mathbf{p}) = f(\mathbf{p}_0) + \sum_{i=1}^{19}\left[\frac{\partial f(\mathbf{p}_0)}{\partial p_i}\right](p_i - p_{i0}) \tag{71}$$

If the $p_i$ are uncorrelated, the performance measure variance $V(f)$ can be estimated as follows based on Equation 71

$$V(f) = \sum_{i=1}^{19}\left[\frac{\partial f(\mathbf{p}_0)}{\partial p_i}\right]^2 V(p_i) \tag{72}$$

where, $V(p_i)$ is the variance of parameter $p_i$.

Equation 72 indicates that if the RN flux is approximated by Equation 71 then the contribution of $p_i$ to the variance of $f$ can be estimated with the following ratio which we shall call the sensitivity index

$$S_i = \left[\frac{\partial f(\mathbf{p}_0)}{\partial p_i}\right]^2 \frac{V(p_i)}{V(f)} \tag{73}$$

The fractional contributions to the variance shown in Equation 73 can be used to order the parameters $p_i$ with respect to their contribution to the uncertainty in $f$. The ordering represented by $S_i$ involves both the absolute effect of the $p_i$, as measured with their partial derivatives, and the effect of distribution assigned to the $p_i$, as measured by $V(f)$. Note that the sensitivity indices take values between 0 and 1 and they all sum up to 1.

The largest computational effort in calculated the sensitivity indices is usually the determination of the partial derivatives appearing in Equations 72 and 73. With the adjoint state method developed here, the computation of all sensitivity indices $S_i$ required the resolution of only one additional system of the form (Equation 41). The variances $V(p_i)$ are estimated using log-normal distributions with the geometric mean corresponding to the reference value and the $2\sigma$ values to extreme values. This approach has been successfully applied in RN release models for performance assessment studies (see for instance Mckinley & Scholits, 1993 and Nancarrow et al., 1988). The extreme values of effective diffusion coefficients and sorption coefficients are taken from Table 4 (columns 8 and 11) of Hayek et al. (2019). The extreme value of the fracture transmissivity is assumed to be equal to $2 \times 10^{-10}$ m$^2$/s.

Figure 9 shows the pie chart percentage of the sensitivity indices calculated from Equation 73. The pie chart indicates that the lateral effective diffusion of the geological formation BD-TA-STA, the fracture transmissivity, and the lateral effective diffusion coefficient of the host rock OPA are the most important parameters which represent all together around two-thirds of the total variance of the RN flux. These three parameters are followed by two other less important parameters that are the sorption coefficient of layer BD-TA-STA and that of the host rock OPA. The remaining parameters are considered as unimportant since each one represents less than 5% of the contribution to the total variance of the RN flux.



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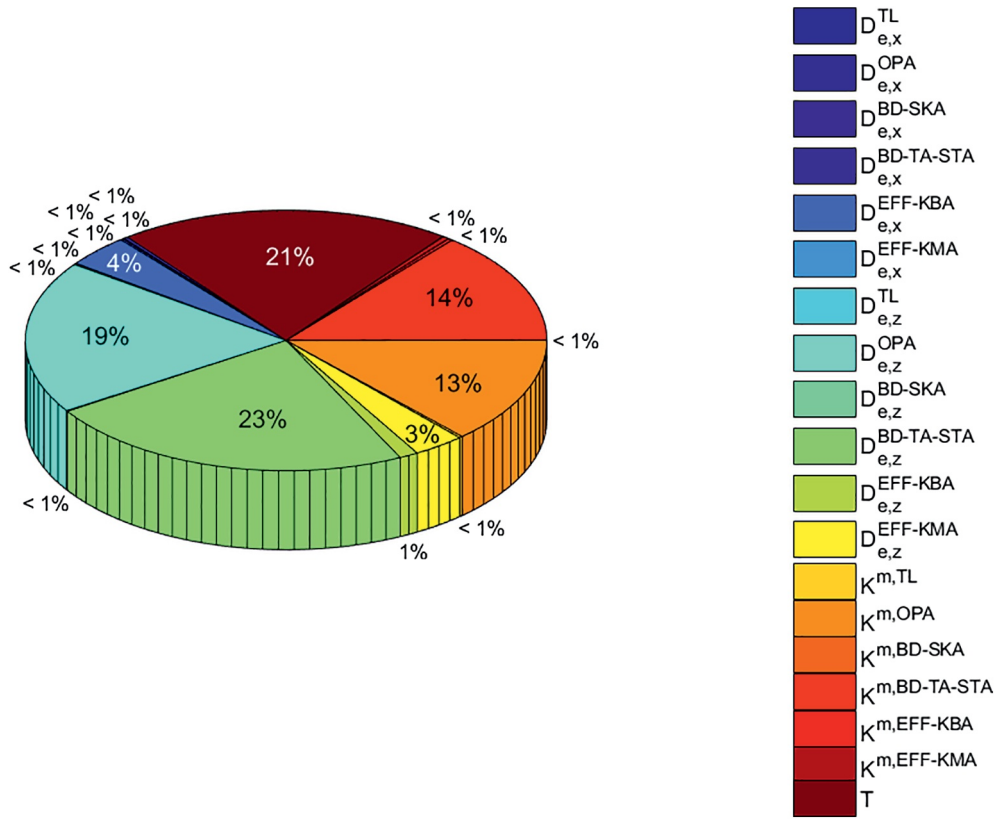



**Figure 9.** Pie chart percentage of the RN flux variance decomposition with respect to the 19 parameters.

The host rock parameters (i.e., the effective diffusion coefficient and the sorption coefficient) were among the important parameters identified in the case of layered formation without fracture investigated in our recent work (Hayek et al., 2019). The fracture transmissivity is an important parameter with respect to the total RN flux at the top because transport in the fracture is dominated by advection and the flux at the top of the model is dominated by the leaving flux from the fracture (i.e., the numerical results have shown that the leaving flux from the fracture is three orders of magnitude larger than that from the porous matrix). The high value of sensitivity index associated with the effective diffusion coefficient of layer BD-TA-STA can be explained by the nominal value of the effective diffusion coefficient of this layer which is relatively higher than all other layers.

It should be mentioned that the results obtained using the DSA method depend on the choice of the base value around which the Taylor series (Equation 71) is developed (i.e., a different base value vector may result in different results). A partial solution of this problem is to construct Taylor series at a number of vectors of the parameter space and then assemble the individual series into a single model. However, we believe that the fracture transmissivity as well as the host rock parameters will remain part of the important parameters for any parameter base value vector around which the Taylor series is expanded.

## 7. Conclusion

The adjoint state methodology applied to the transport of radionuclides in a heterogeneous porous media including a fracture in steady-state regime has been presented. The system of governing equations is composed of two PDEs: the first representing 1D transport in the fracture which is characterized by its high transmissivity, and the second is used to model 2D transport in the rock matrix. The two equations are connected by a flux term representing the exchange between the two media. The adjoint state method aims to compute, in an efficient computational way, the sensitivity coefficients (which are the main tools of sensitivity analysis) of a predefined performance measure with respect to the transport parameters.



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Both continuous and discrete adjoint approaches have been employed. Using the continuous approach, we derived the system of adjoint state equations in terms of PDEs. This system is comparable to the system of transport equations in the sense that it is composed of two adjoint equations, one associated with the fracture and one associated with the rock matrix with a flux term connecting the two equations. Each adjoint state equation contains a load (source) term which depends on the performance measure. 14 general formulas of sensitivity coefficients associated with all types of parameters (including transport parameters, boundary conditions and fracture aperture) have been formulated in integral forms. We derived exact analytical solutions in heterogeneous layered medium for both transport and adjoint systems associated with a specific performance measure defined as concentration integral. These analytical solutions can be considered as reference solutions to validate any numerical adjoint model. The analytical results have been compared with developed numerical results and have shown excellent agreements.

The discrete adjoint approach, which is based on the numerical discretized form of the transport system, has been coupled with the differential sensitivity analysis method and applied to a real field case involving the performance assessment of a site currently being considered for underground nuclear storage in northern Switzerland, Zürich Nordost. The numerical model involved 19 parameters including the effective diffusion coefficients and the sorption coefficients of six geological formations as well as the transmissivity of a fracture crossing the waste repository. The results show that fracture transmissivity is among the most important parameters with respect to the released flux to the biosphere and then plays an essential role in the safety assessment of radioactive waste disposal.

We note that the results presented here apply to the steady-state case and can be readily extended to the time-dependent case. The main difference between the two cases is the presence of the time derivative operators in the governing transport equations of the time-depend case and the need of defining initial conditions. Therefore, the general expression of the performance measure (Equation 20) would involve integral with respect to the time variable. This would lead to a generalization of the whole procedure presented in Appendix A to derive the time-dependent adjoint equations involving integration by part with respect to the time variable. Note that the time-dependent case has some unique feature that must be treated differently from the steady-state case. For instance, time-dependent performance measures (e.g., maximum dose rate of RNs over time at a predefined interface referred to as the accessible environment) are important for safety assessment. The adjoint state sensitivity method applied to the transient case will be the subject of a forthcoming work.

## Appendix A: Derivation of Adjoint State Equations and Sensitivity Coefficients

To derive the adjoint state equations associated with Equations 12 and 13 and the marginal sensitivity coefficient (Equation 34), we take the first derivatives of Equations 12 and 13 with respect to a given parameter $p$. We get:

$$\begin{aligned}\frac{\partial L^f}{\partial p} \equiv & \frac{\mathrm{d}}{\mathrm{d}x}\left(-\frac{\partial D_e^f}{\partial p}\frac{\mathrm{d}C^f}{\mathrm{d}x}\right)+\frac{\mathrm{d}}{\mathrm{d}x}\left(-D_e^f\frac{\mathrm{d}\psi^f}{\mathrm{d}x}\right)+\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial V^f}{\partial p}C^f\right)+\frac{\mathrm{d}}{\mathrm{d}x}\left(V^f\psi^f\right)+\lambda\frac{\partial\phi^f}{\partial p}R^fC^f \\ & +\lambda\phi^f\frac{\partial R^f}{\partial p}C^f+\lambda\phi^fR^f\psi^f-\frac{\partial}{\partial p}\left(\frac{D_{e,z}^m}{b}\frac{\partial C^m}{\partial z}\Big|_{z=b}\right)+\frac{\partial Q^f}{\partial p}+\frac{\partial Q^f}{\partial C^f}\psi^f=0\end{aligned} \tag{A1}$$

and

$$\begin{aligned}\frac{\partial L^m}{\partial p} \equiv & \frac{\partial}{\partial x}\left(-\frac{\partial D_{e,x}^m}{\partial p}\frac{\partial C^m}{\partial x}\right)+\frac{\partial}{\partial x}\left(-D_{e,x}^m\frac{\partial\psi^m}{\partial x}\right)+\frac{\partial}{\partial z}\left(-\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\right)+\frac{\partial}{\partial z}\left(-D_{e,z}^m\frac{\partial\psi^m}{\partial z}\right) \\ & +\frac{\partial}{\partial x}\left(\frac{\partial V^m}{\partial p}C^m\right)+\frac{\partial}{\partial x}\left(V^m\psi^m\right)+\lambda\frac{\partial\phi^m}{\partial p}R^mC^m+\lambda\phi^m\frac{\partial R^m}{\partial p}C^m+\lambda\phi^mR^m\psi^m \\ & +\frac{\partial Q^m}{\partial p}+\frac{\partial Q^m}{\partial C^m}\psi^m=0\end{aligned} \tag{A2}$$

The boundary conditions associated with Equations A1 and A2 can be obtained by taking the derivatives of Equations 14–19 with respect to $p$:

For the fracture:

$$\psi^f\left(0\right) = \frac{\partial C_0^f}{\partial p}, \quad 0 \le z \le b \tag{A3}$$

$$\psi^f\left(L\right) = \frac{\partial C_L^f}{\partial p}, \quad 0 \le z \le b \tag{A4}$$

For the matrix:

$$\psi^m\left(0,z\right) = \frac{\partial C_0^m}{\partial p}, \quad b \le z \le B \tag{A5}$$

$$\psi^m\left(L,z\right) = \frac{\partial C_L^m}{\partial p}, \quad b \le z \le B \tag{A6}$$

$$\psi^m\left(x,b\right) = \psi^f\left(x\right), \quad 0 \le x \le L \tag{A7}$$

$$\frac{\partial \psi^m}{\partial z}\Big|_{z=B} = 0, \quad 0 \le x \le L \tag{A8}$$

To compute the integral defined by Equation 22 (resp. Equation 23), we multiply each term of the RHS of Equation A1 (resp. Equation A2) by $\psi_*^f$ (resp. $\psi_*^m$) and we intergrate over $\Omega^f$ (resp. $\Omega^m$). The integrals can be computed one by one using the partial integration rule as follows.

The integral $\iint_{\Omega^f} \mathrm{d}/\mathrm{d}x\left(-\partial D_e^f/\partial p \mathrm{d}C^f/\mathrm{d}x\right)\psi_*^f d\Omega$ can be calculated as follows:

$$\begin{aligned}\iint_{\Omega^f} \frac{\mathrm{d}}{\mathrm{d}x}\left(-\frac{\partial D_e^f}{\partial p}\frac{\mathrm{d}C^f}{\mathrm{d}x}\right)\psi_*^f d\Omega &= \int_0^L\int_0^b \frac{\mathrm{d}}{\mathrm{d}x}\left(-\frac{\partial D_e^f}{\partial p}\frac{\mathrm{d}C^f}{\mathrm{d}x}\right)\psi_*^f d\Omega \\ &= \int_0^b\left\{\left[-\frac{\partial D_e^f}{\partial p}\frac{\mathrm{d}C^f}{\mathrm{d}x}\psi_*^f\right]_{x=0}^{x=L} + \int_0^L \frac{\partial D_e^f}{\partial p}\frac{\mathrm{d}C^f}{\mathrm{d}x}\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\mathrm{d}x\right\}\mathrm{d}z \\ &= \iint_{\Omega^f} \frac{\partial D_e^f}{\partial p}\frac{\mathrm{d}C^f}{\mathrm{d}x}\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}d\Omega\end{aligned} \tag{A9}$$

The first term of the second line of Equation A9 has been eliminated by imposing the following boundary conditions on $\psi_*^f$

$$\psi_*^f\left(0\right) = 0, \quad 0 \le z \le b \tag{A10}$$

$$\psi_*^f\left(L\right) = 0, \quad 0 \le z \le b \tag{A11}$$

The integral $\iint_{\Omega^f} \mathrm{d}/\mathrm{d}x\left(-D_e^f \mathrm{d}\psi^f/\mathrm{d}x\right)\psi_*^f d\Omega$ can be calculated as follows:

$$\begin{aligned}\iint_{\Omega^f} \frac{\mathrm{d}}{\mathrm{d}x}\left(-D_e^f\frac{\mathrm{d}\psi^f}{\mathrm{d}x}\right)\psi_*^f d\Omega &= \int_0^L\int_0^b \frac{\mathrm{d}}{\mathrm{d}x}\left(-D_e^f\frac{\mathrm{d}\psi^f}{\mathrm{d}x}\right)\psi_*^f d\Omega \\ &= \int_0^b\left\{\left[-D_e^f\frac{\mathrm{d}\psi^f}{\mathrm{d}x}\psi_*^f\right]_{x=0}^{x=L} + \int_0^L D_e^f\frac{\mathrm{d}\psi^f}{\mathrm{d}x}\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\mathrm{d}x\right\}\mathrm{d}z \\ &= \int_0^b\left\{\left[D_e^f\psi^f\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\right]_{x=0}^{x=L} + \int_0^L \frac{\mathrm{d}}{\mathrm{d}x}\left(-D_e^f\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\right)\psi^f\mathrm{d}x\right\}\mathrm{d}z \\ &= \iint_{\Omega^f}\frac{D_e^f}{L}\left(\frac{\partial C_L^f}{\partial p}\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\Big|_{x=L} - \frac{\partial C_0^f}{\partial p}\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\Big|_{x=0}\right)d\Omega + \iint_{\Omega^f}\frac{\mathrm{d}}{\mathrm{d}x}\left(-D_e^f\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\right)\psi^f d\Omega\end{aligned} \tag{A12}$$

The first term of the second line of Equation A12 has been eliminated by using the imposed conditions A10 and A11, and in the fourth term of Equation A12 we have used the boundary conditions A3 and A4.



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The integral $\iint_{\Omega^f} \mathrm{d}/\mathrm{d}x\left(\partial V^f / \partial p C^f\right)\psi_*^f d\Omega$ can be calculated as follows:

$$\begin{aligned}\iint_{\Omega^f} \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial V^f}{\partial p}C^f\right)\psi_*^f d\Omega &= \int_0^L\int_0^b \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial V^f}{\partial p}C^f\right)\psi_*^f d\Omega \\ &= \int_0^b\left\{\left[\frac{\partial V^f}{\partial p}C^f\psi_*^f\right]_{x=0}^{x=L} - \int_0^L \frac{\partial V^f}{\partial p}C^f\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\mathrm{d}x\right\}\mathrm{d}z \\ &= -\iint_{\Omega^f}\frac{\partial V^f}{\partial p}C^f\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}d\Omega\end{aligned} \tag{A13}$$

The first term of the second line of Equation A13 has been eliminated by using the imposed conditions A10 and A11.

The integral $\iint_{\Omega^f}\frac{\mathrm{d}}{\mathrm{d}x}\left(V^f\psi^f\right)\psi_*^f d\Omega$ can be calculated as follows:

$$\begin{aligned}\iint_{\Omega^f}\frac{\mathrm{d}}{\mathrm{d}x}\left(V^f\psi^f\right)\psi_*^f d\Omega &= \int_0^L\int_0^b\frac{\mathrm{d}}{\mathrm{d}x}\left(V^f\psi^f\right)\psi_*^f d\Omega \\ &= \int_0^b\left\{\left[V^f\psi^f\psi_*^f\right]_{x=0}^{x=L} - \int_0^L V^f\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\psi^f\mathrm{d}x\right\}\mathrm{d}z \\ &= -\iint_{\Omega^f}V^f\frac{\mathrm{d}\psi_*^f}{\mathrm{d}x}\psi^f d\Omega\end{aligned} \tag{A14}$$

The first term of the second line of Equation A14 has been eliminated by using the imposed conditions A10 and A11.

To calculate the integral $\iint_{\Omega^f}\partial / \partial p\left(D_{e,z}^m / b\partial C^m / \partial z\mid_{z=b}\right)\psi_*^f d\Omega$ we first expand the term $\partial/\partial p\left(D_{e,z}^m / b\partial C^m/\partial z\mid_{z=b}\right)$ in the integrand as follows

$$\begin{aligned}\frac{\partial}{\partial p}\left(\frac{D_{e,z}^m}{b}\frac{\partial C^m}{\partial z}\Big|_{z=b}\right) &= \frac{1}{b}\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\Big|_{z=b} - \frac{D_{e,z}^m}{b^2}\frac{\partial b}{\partial p}\frac{\partial C^m}{\partial z}\Big|_{z=b} + \frac{D_{e,z}^m}{b}\frac{\partial}{\partial p}\left(\frac{\partial C^m}{\partial z}\Big|_{z=b}\right) \\ &= \frac{1}{b}\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\Big|_{z=b} - \frac{D_{e,z}^m}{b^2}\frac{\partial b}{\partial p}\frac{\partial C^m}{\partial z}\Big|_{z=b} + \frac{D_{e,z}^m}{b}\frac{\partial \psi^m}{\partial z}\Big|_{z=b} + \frac{D_{e,z}^m}{b}\frac{\partial b}{\partial p}\frac{\partial^2 C^m}{\partial z^2}\Big|_{z=b}\end{aligned} \tag{A15}$$

Therefore, we get

$$\begin{aligned}\iint_{\Omega^f}\frac{\partial}{\partial p}\left(\frac{D_{e,z}^m}{b}\frac{\partial C^m}{\partial z}\Big|_{z=b}\right)\psi_*^f d\Omega = &\iint_{\Omega^f}\frac{\partial D_{e,z}^m}{\partial p}\frac{1}{b}\frac{\partial C^m}{\partial z}\Big|_{z=b}\psi_*^f d\Omega + \iint_{\Omega^f}\frac{D_{e,z}^m}{b}\frac{\partial \psi^m}{\partial z}\Big|_{z=b}\psi_*^f d\Omega \\ &-\iint_{\Omega^f}\frac{\partial b}{\partial p}\frac{D_{e,z}^m}{b}\left(\frac{1}{b}\frac{\partial C^m}{\partial z}\Big|_{z=b} - \frac{\partial^2 C^m}{\partial z^2}\Big|_{z=b}\right)\psi_*^f d\Omega\end{aligned} \tag{A16}$$

In a similar way, we calculate the terms of the integral defined by Equation 23 as follows.

The integral $\iint_{\Omega^m}\partial / \partial x\left(-\partial D_{e,x}^m / \partial p\partial C^m / \partial x\right)\psi_*^m d\Omega$ can be calculated as follows:

$$\begin{aligned}\iint_{\Omega^m}\frac{\partial}{\partial x}\left(-\frac{\partial D_{e,x}^m}{\partial p}\frac{\partial C^m}{\partial x}\right)\psi_*^m d\Omega &= \int_0^L\int_b^B\frac{\partial}{\partial x}\left(-\frac{\partial D_{e,x}^m}{\partial p}\frac{\partial C^m}{\partial x}\right)\psi_*^m d\Omega \\ &= \int_b^B\left\{\left[-\frac{\partial D_{e,x}^m}{\partial p}\frac{\partial C^m}{\partial x}\psi_*^m\right]_{x=0}^{x=L} + \int_0^L\frac{\partial D_{e,x}^m}{\partial p}\frac{\partial C^m}{\partial x}\frac{\partial \psi_*^m}{\partial x}\mathrm{d}x\right\}\mathrm{d}z \\ &= \iint_{\Omega^m}\frac{\partial D_{e,x}^m}{\partial p}\frac{\partial C^m}{\partial x}\frac{\partial \psi_*^m}{\partial x}d\Omega\end{aligned} \tag{A17}$$

The first term of the second line of Equation A17 has been eliminated by imposing the following boundary conditions on $\psi_*^m$



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$$\psi_*^m\left(0,z\right)=0,\quad b\le z\le B \tag{A18}$$

$$\psi_*^m\left(L,z\right)=0,\quad b\le z\le B \tag{A19}$$

The integral $\iint_{\Omega^m}\partial\,/\,\partial x\left(-D_{e,x}^m\partial\psi^m\,/\,\partial x\right)\psi_*^m d\Omega$ can be calculated as follows:

$$\begin{aligned}\iint_{\Omega^m}\frac{\partial}{\partial x}\left(-D_{e,x}^m\frac{\partial\psi^m}{\partial x}\right)\psi_*^m d\Omega&=\int_0^L\int_b^B\frac{\partial}{\partial x}\left(-D_{e,x}^m\frac{\partial\psi^m}{\partial x}\right)\psi_*^m d\Omega\\&=\int_b^B\left\{\left[-D_{e,x}^m\frac{\partial\psi^m}{\partial x}\psi_*^m\right]_{x=0}^{x=L}+\int_0^L D_{e,x}^m\frac{\partial\psi^m}{\partial x}\frac{\partial\psi_*^m}{\partial x}\mathrm{d}x\right\}\mathrm{d}z\\&=\int_b^B\left\{\left[D_{e,x}^m\psi^m\frac{\partial\psi_*^m}{\partial x}\right]_{x=0}^{x=L}+\int_0^L\frac{\partial}{\partial x}\left(-D_{e,x}^m\frac{\partial\psi_*^m}{\partial x}\right)\psi^m\mathrm{d}x\right\}\mathrm{d}z\\&=\iint_{\Omega^m}\frac{D_{e,x}^m}{L}\left(\frac{\partial C_L^m}{\partial p}\frac{\partial\psi_*^m}{\partial x}\Big|_{x=L}-\frac{\partial C_0^m}{\partial p}\frac{\partial\psi_*^m}{\partial x}\Big|_{x=0}\right)d\Omega\\&\quad+\iint_{\Omega^m}\frac{\partial}{\partial x}\left(-D_{e,x}^m\frac{\partial\psi_*^m}{\partial x}\right)\psi^m d\Omega\end{aligned} \tag{A20}$$

The first term of the second line of Equation A20 has been eliminated by using the imposed conditions A18 and A19, and in the fourth term of Equation A20 we have used the boundary conditions A5 and A6.

The integral $\iint_{\Omega^m}\partial/\partial z\left(-\partial D_{e,z}^m/\partial p\partial C^m/\partial z\right)\psi_*^m d\Omega$ can be calculated as follows:

$$\begin{aligned}\iint_{\Omega^m}\frac{\partial}{\partial z}\left(-\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\right)\psi_*^m d\Omega&=\int_0^L\int_b^B\frac{\partial}{\partial z}\left(-\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\right)\psi_*^m\,d\Omega\\&=\int_0^L\left\{\left[-\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\psi_*^m\right]_{z=b}^{z=B}+\int_b^B\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\frac{\partial\psi_*^m}{\partial z}\mathrm{d}z\right\}\mathrm{d}x\\&=\int_0^L\left\{\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\Big|_{z=b}\psi_*^m\left(x,b\right)+\int_b^B\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\frac{\partial\psi_*^m}{\partial z}\mathrm{d}z\right\}\mathrm{d}x\\&=\iint_{\Omega^f}\frac{\partial D_{e,z}^m}{\partial p}\frac{1}{b}\frac{\partial C^m}{\partial z}\Big|_{z=b}\psi_*^f d\Omega+\iint_{\Omega^m}\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\frac{\partial\psi_*^m}{\partial z}d\Omega\end{aligned} \tag{A21}$$

In the third line of Equation A21 we have used the boundary condition 19. The fourth line of Equation A21 is deduced from the third line by imposing the following condition at the interface between the fracture and rock matrix:

$$\psi_*^m\left(x,b\right)=\psi_*^f\left(x\right),\quad 0\le x\le L \tag{A22}$$

The integral $\iint_{\Omega^m}\partial\,/\,\partial z\left(-D_{e,z}^m\partial\psi^m\,/\,\partial z\right)\psi_*^m d\Omega$ can be calculated as follows:

$$\begin{aligned}\iint_{\Omega^m}\frac{\partial}{\partial z}\left(-D_{e,z}^m\frac{\partial\psi^m}{\partial z}\right)\psi_*^m d\Omega&=\int_0^L\int_b^B\frac{\partial}{\partial z}\left(-D_{e,z}^m\frac{\partial\psi^m}{\partial z}\right)\psi_*^m\,d\Omega\\&=\int_0^L\left\{\left[-D_{e,z}^m\frac{\partial\psi^m}{\partial z}\psi_*^m\right]_{z=b}^{z=B}+\int_b^B D_{e,z}^m\frac{\partial\psi^m}{\partial z}\frac{\partial\psi_*^m}{\partial z}\mathrm{d}z\right\}\mathrm{d}x\\&=\int_0^L\left\{D_{e,z}^m\frac{\partial\psi^m}{\partial z}\Big|_{z=b}\psi_*^m\left(x,b\right)+\int_b^B D_{e,z}^m\frac{\partial\psi^m}{\partial z}\frac{\partial\psi_*^m}{\partial z}\mathrm{d}z\right\}\mathrm{d}x\\&=\iint_{\Omega^f}\frac{D_{e,z}^m}{b}\frac{\partial\psi^m}{\partial z}\Big|_{z=b}\psi_*^f d\Omega+\iint_{\Omega^m}\frac{\partial D_{e,z}^m}{\partial p}\frac{\partial C^m}{\partial z}\frac{\partial\psi_*^m}{\partial z}d\Omega\end{aligned} \tag{A23}$$



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In the third line of Equation A23 we have imposed the following boundary condition on $\psi^m$:

$$\frac{\partial \psi_*^m}{\partial z}\Big|_{z=B} = 0, \quad 0 \le x \le L \tag{A24}$$

The fourth line of Equation A23 is obtained from the third line by using the interface condition A22.

The integral $\iint_{\Omega^m} \partial / \partial x \left( \partial V^m / \partial p C^m \right) \psi_*^m d\Omega$ can be calculated as follows:

$$\begin{aligned} \iint_{\Omega^m} \left( \frac{\partial V^m}{\partial p} C^m \right) \psi_*^m d\Omega &= \int_0^L \int_b^B \frac{\partial}{\partial x} \left( \frac{\partial V^m}{\partial p} C^m \right) \psi_*^m d\Omega \\ &= \int_b^B \left\{ \left[ \frac{\partial V^m}{\partial p} C^m \psi_*^m \right]_{x=0}^{x=L} - \int_0^L \frac{\partial V^m}{\partial p} C^m \frac{\partial \psi_*^m}{\partial x} \mathrm{d}x \right\} \mathrm{d}z \\ &= -\iint_{\Omega^m} \frac{\partial V^m}{\partial p} C^m \frac{\partial \psi_*^m}{\partial x} d\Omega \end{aligned} \tag{A25}$$

The first term of the second line of Equation A13 has been eliminated by using the imposed conditions A18 and A19.

The integral $\iint_{\Omega^m} \partial / \partial x \left( V^m \psi^m \right) \psi_*^m d\Omega$ can be calculated as follows:

$$\begin{aligned} \iint_{\Omega^m} \frac{\partial}{\partial x} \left( V^m \psi^m \right) \psi_*^m d\Omega &= \int_0^L \int_b^B \frac{\partial}{\partial x} \left( V^m \psi^m \right) \psi_*^m d\Omega \\ &= \int_b^B \left\{ \left[ V^m \psi^m \psi_*^m \right]_{x=0}^{x=L} - \int_0^L V^m \frac{\partial \psi_*^m}{\partial x} \psi^m \mathrm{d}x \right\} \mathrm{d}z \\ &= -\iint_{\Omega^m} V^m \frac{\partial \psi_*^m}{\partial x} \psi^m d\Omega \end{aligned} \tag{A26}$$

The first term of the second line of Equation A26 has been eliminated by using the imposed conditions A18 and A19.

## Appendix B: Derivation of Exact Solutions

The analytical expression of the concentration in the $i^{th}$ layer of the rock matrix can be obtained by solving Equation 48 subject to boundary conditions 51 and 52. The general form solution can be written as follows:

$$C_i^m(x,z) = [C_i^f(x) + \frac{B_i^m}{\lambda}] \frac{\cosh[r_i^m (B-z)]}{\cosh[r_i^m (B-b_i)]} - \frac{B_i^m}{\lambda} \tag{B1}$$

with

$$B_i^m = \frac{Q_i^m}{\phi_i^m R_i^m} \tag{B2}$$

$$r_i^m = c_{ri}^m \lambda^{1/2} \tag{B3}$$

$$c_{ri}^m = \left( \frac{\phi_i^m R_i^m}{D_{e,z,i}^m} \right)^{1/2} \tag{B4}$$

The substitution of Equation B1 into Equation 47 yields

$$-D_{e,i}^f \frac{\mathrm{d}^2 C_i^f}{\mathrm{d}x^2} + V_i^f \frac{\mathrm{d}C_i^f}{\mathrm{d}x} + r_{ai} C_i^f = -Q_i^f - \frac{c_{fi} B_i^m}{\lambda^{1/2}} \tag{B5}$$

with

$$c_{fi} = \frac{\left( \phi_i^m R_i^m D_{e,z,i}^m \right)^{1/2}}{b_i} \tanh[r_i^m (B - b_i)] \tag{B6}$$

$$r_{ai} = \lambda \phi_i^f R_i^f + \lambda^{1/2} c_{fi} \tag{B7}$$

The general form solution of Equation B5 can be written as follows:



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$$C_i^f(x) = e^{\nu_i x}\left[A_i \sinh(\delta_i x) + B_i \cosh(\delta_i x)\right] + F_i \tag{B8}$$

with

$$\nu_i = \frac{V_i^f}{2D_{e,i}^f} \tag{B9}$$

$$\delta_i = \left(\nu_i^2 + \frac{r_{ai}}{D_{e,i}^f}\right)^{1/2} \tag{B10}$$

$$F_i = -\frac{Q_i^f}{r_{ai}} - \frac{c_{fi}B_i^m}{\lambda^{1/2} r_{ai}} \tag{B11}$$

where $A_i$ and $B_i$ are two constants which can be determined by using two physical conditions which must be satisfied at the interface between two layers that are the continuity of concentration (Equation B12) and the continuity of flux (Equation B13)

$$C_i^f(x_i) = C_{i+1}^f(x_i) \tag{B12}$$

$$F_i(x_i) = F_{i+1}(x_i) \tag{B13}$$

where $F_i$ is the fracture advective/dispersive flux in layer $i$ defined by

$$F_i(x) = b_i\left[-D_{e,i}^f \frac{dC_i^f}{dx}(x) + V_i^f C_i^f(x)\right] \tag{B14}$$

Taking into account the continuity of fluid flux in the fracture along interfaces between layers (i.e., $b_i V_i^f = b_{i+1} V_{i+1}^f$) and the continuity of concentration (Equation B12), Equation B13 reduces to

$$b_i D_{e,i}^f \frac{dC_i^f}{dx}(x_i) = b_{i+1} D_{e,i+1}^f \frac{dC_{i+1}^f}{dx}(x_i) \tag{B15}$$

Equation B12 indicates that at the interface $x = x_i$, there is one single value of concentration (no jump of concentration at the interface). We shall denote this value by $C_i$ (i.e., $C_i = C_i^f(x_i) = C_{i+1}^f(x_i)$). Substituting $C_i^f(x_{i-1}) = C_{i-1}$ and $C_i^f(x_i) = C_i$ into Equation B8, we get the following linear algebraic system of two equations with two unknowns $A_i$ and $B_i$:

$$\left[\sinh(\delta_i x_{i-1})\right]A_i + \left[\cosh(\delta_i x_{i-1})\right]B_i = (C_{i-1} - F_i)e^{-\nu_i x_{i-1}} \tag{B16}$$

$$\left[\sinh(\delta_i x_i)\right]A_i + \left[\cosh(\delta_i x_i)\right]B_i = (C_i - F_i)e^{-\nu_i x_i} \tag{B17}$$

Solving system (B16) and (B17) yields $A_i$ and $B_i$ as functions of $C_{i-1}$ and $C_i$ as follows

$$A_i = -(C_{i-1} - F_i)e^{-\nu_i x_{i-1}} \frac{\cosh(\delta_i x_i)}{\sinh(\delta_i L_i)} + (C_i - F_i)e^{-\nu_i x_i} \frac{\cosh(\delta_i x_{i-1})}{\sinh(\delta_i L_i)} \tag{B18}$$

$$B_i = -(C_i - F_i)e^{-\nu_i x_i} \frac{\sinh(\delta_i x_{i-1})}{\sinh(\delta_i L_i)} + (C_{i-1} - F_i)e^{-\nu_i x_{i-1}} \frac{\sinh(\delta_i x_i)}{\sinh(\delta_i L_i)} \tag{B19}$$

The substitution of Equations B18 and B19 into Equation B8 yields for $i = 1, 2, \ldots, N-1$

$$C_i^f(x) = (C_i - F_i)e^{\nu_i(x - x_i)} \frac{\sinh\left[\delta_i(x - x_{i-1})\right]}{\sinh(\delta_i L_i)} + (C_{i-1} - F_i)e^{\nu_i(x - x_{i-1})} \frac{\sinh\left[\delta_i(x_i - x)\right]}{\sinh(\delta_i L_i)} + F_i \tag{B20}$$

where $C_0 = C_0^f$ and $C_N = C_L^f$.

The solution (B20) is not complete yet because $C_i$'s are not known yet. To determine these values we use the flux continuity condition (B15). Using Equations B20 and B15, we get the following relationship between $C_{i-1}$, $C_i$, and $C_{i+1}$

$$d_1 C_1 + c_1 C_2 = f_1 \tag{B21}$$

$$a_i C_{i-1} + d_i C_i + c_i C_{i+1} = f_i, \quad i = 2, \ldots, N-2 \tag{B22}$$



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$$a_{N-1}C_{N-2} + d_{N-1}C_{N-1} = f_{N-1} \tag{B23}$$

with

$$a_i = -\frac{b_i D_{e,i}^f \delta_i e^{v_i L_i}}{\sinh\left(\delta_i L_i\right)}, \quad i = 2,\ldots,N-1 \tag{B24}$$

$$c_i = -\frac{b_{i+1} D_{e,i+1}^f \delta_{i+1} e^{-v_{i+1} L_{i+1}}}{\sinh\left(\delta_{i+1} L_{i+1}\right)}, \quad i = 1,\ldots,N-2 \tag{B25}$$

$$d_i = \frac{b_i D_{e,i}^f \delta_i \cosh\left(\delta_i L_i\right)}{\sinh\left(\delta_i L_i\right)} + \frac{b_{i+1} D_{e,i+1}^f \delta_{i+1} \cosh\left(\delta_{i+1} L_{i+1}\right)}{\sinh\left(\delta_{i+1} L_{i+1}\right)}, \quad i = 1,\ldots,N-1 \tag{B26}$$

$$f_1 = \frac{1}{2} b_1 V_1^f F_1 - \frac{1}{2} b_2 V_2^f F_2 + a_1 F_1 + c_1 F_2 + \frac{b_1 D_{e,1}^f \delta_1 F_1 \cosh\left(\delta_1 L_1\right)}{\sinh\left(\delta_1 L_1\right)} + \frac{b_2 D_{e,2}^f \delta_2 F_2 \cosh\left(\delta_2 L_2\right)}{\sinh\left(\delta_2 L_2\right)} + \frac{b_1 D_{e,1}^f \delta_1 e^{v_1 L_1}}{\sinh\left(\delta_1 L_1\right)} C_0^f \tag{B27}$$

$$f_i = \frac{1}{2} b_i V_i^f F_i - \frac{1}{2} b_{i+1} V_{i+1}^f F_{i+1} + a_i F_i + \frac{b_i D_{e,i}^f \delta_i F_i \cosh\left(\delta_i L_i\right)}{\sinh\left(\delta_i L_i\right)} + c_i F_{i+1} + \frac{b_{i+1} D_{e,i+1}^f \delta_{i+1} F_{i+1} \cosh\left(\delta_{i+1} L_{i+1}\right)}{\sinh\left(\delta_{i+1} L_{i+1}\right)}, \quad i = 2,\&,N-2 \tag{B28}$$

$$f_{N-1} = \frac{1}{2} b_{N-1} V_{N-1}^f F_{N-1} - \frac{1}{2} b_N V_N^f F_N + a_{N-1} F_{N-1} + \frac{b_{N-1} D_{e,N-1}^f \delta_{N-1} F_{N-1} \cosh\left(\delta_{N-1} L_{N-1}\right)}{\sinh\left(\delta_{N-1} L_{N-1}\right)} + c_{N-1} F_N + \frac{b_N D_{e,N}^f \delta_N F_N \cosh\left(\delta_N L_N\right)}{\sinh\left(\delta_N L_N\right)} + \frac{b_N D_{e,N}^f \delta_N e^{-v_N L_N}}{\sinh\left(\delta_N L_N\right)} C_L^f \tag{B29}$$

The system of Equations B21–B23 can be written is matrix form as follows

$$[A]\{C\} = \{F\} \tag{B30}$$

with

$$[A] = \begin{bmatrix} d_1 & c_1 & & & & & \\ a_2 & d_2 & c_2 & & & & \\ & \ddots & \ddots & \ddots & & & \\ & & a_j & d_j & c_j & & \\ & & & \ddots & \ddots & \ddots & \\ & & & & a_{N-2} & d_{N-2} & c_{N-2} \\ & & & & & a_{N-1} & d_{N-1} \end{bmatrix} \tag{B31}$$

and

$$\{C\} = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_{N-1} \end{bmatrix}, \quad \{F\} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_{N-1} \end{bmatrix} \tag{B32}$$

The solution of system B30 can be written then as follows

$$\{C\} = [A^{-1}]\{F\} \tag{B33}$$

where $[A^{-1}]$ is the inverse of marix $[A]$.



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The matrix $[A]$ is a tridiagonal matrix. Its inverse $[A^{-1}]$ can be obtained analytically based on the coefficients of the matrix $[A]$. If we denote $[A^{-1}] = \{\varphi_{i,j}\}\,(1 \le i, j \le N-1)$, the coefficients $\varphi_{i,j}$ can be expresed as follows (Huang & McColl, 1997)

$$\varphi_{j,j} = \frac{1}{d_j - a_j c_{j-1}\dfrac{z_{j-2}}{z_{j-1}} - a_{j+1}c_j\dfrac{y_{j+2}}{y_{j+1}}} \tag{B34}$$

where $j = 1,2,\ldots,N-1$, $a_1 = 0$, $c_{N-1} = 0$, and

$$\varphi_{i,j} = \begin{cases} (-1)^{j-i}\left(\prod_{k=1}^{j-i} c_{j-k}\right)\dfrac{z_{i-1}}{z_{j-1}}\varphi_{j,j}, & i < j \\ (-1)^{i-j}\left(\prod_{k=1}^{i-j} a_{j+k}\right)\dfrac{y_{j+1}}{y_{j+1}}\varphi_{j,j}, & i > j \end{cases} \tag{B35}$$

In Equations B34 and B35 $\{z_i\}$ and $\{y_i\}$ are second-order linear recurrences defined by

$$z_i = d_i z_{i-1} - a_i c_{i-1} z_{i-2}, \quad i = 2,3,\ldots,N-1 \tag{B36}$$

where $z_0 = 1$, $z_1 = d_1$, and

$$y_j = d_j y_{j+1} - a_{j+1}c_j y_{j+2}, \quad j = N-2, N-3,\ldots,1 \tag{B37}$$

where $y_N = 1$, $y_{N-1} = d_{N-1}$.

Using Equation B33, the analytical expression of $C_i$ for $i = 1,2,\ldots,N-1$ can be written as

$$C_i = \sum_{j=1}^{N-1} \varphi_{i,j} f_j \tag{B38}$$

## Data Availability Statement

Data used for the validation of analytical results are available through Gureghian et al. (1993). Data used in the application to the real field case is available through Hayek et al. (2019) and Nagra (2014).

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